SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Design and Control of an Aerodynamic Surface-Enhanced Multirotor

William Constantin Rosenhahn

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Author:	William Constantin Rosenhahn
Examiner:	Prof. Dr.-Ing. Markus Ryll
Supervisor:	Lukas Pries
Submission Date:	01.12.2025

I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 01.12.2025 William Constantin Rosenhahn

Acknowledgments

"I would like to thank Pablo Kirby for developing the improved platform variant and supporting the experimental validation, and for the collaborative work environment."

Abstract

This thesis presents the design, modeling, and experimental validation of an X-wing hybrid aerial platform that combines quadrotor vertical take-off and landing (VTOL) capability with fixed-wing aerodynamic efficiency. The platform features four wings arranged in an X-configuration at 45° dihedral, providing passive lift augmentation during forward flight while maintaining full quadrotor control authority.

A comprehensive dynamics model incorporating aerodynamic forces and moments is developed and integrated with an Incremental Nonlinear Dynamic Inversion (INDI) control architecture. The controller requires no explicit aerodynamic model and treats wing-generated forces as bounded disturbances, enabling robust trajectory tracking across the entire flight envelope from hover to high-speed cruise.

Experimental validation was conducted in a motion capture arena with a 2.5 kg prototype featuring NACA 0015 airfoils. Thrust characterization yielded the mapping $T = 5.15 \times 10^{-6} \cdot \mathrm{RPM}^2 - 0.090 \times 10^{-2} \cdot \mathrm{RPM}$ (N) with $R^2 = 0.997$. Flight tests demonstrated stable, accurate tracking up to $8\,\mathrm{m\,s^{-1}}$ with RMS position errors between 0.03 m and 0.16 m.

Aerodynamic coefficient identification from flight data revealed that the effective wing area is approximately 50% of the geometric area due to fuselage blockage and interference effects. At $9\,\mathrm{m\,s^{-1}}$ cruise speed, the wings reduced rotor power requirements by approximately 12% compared to conventional quadrotor flight. An optimized AG25 airfoil variant achieved 50% power savings at the same cruise condition, demonstrating the potential for substantial efficiency gains through aerodynamic optimization.

The results verify that INDI can maintain precise control on an aerodynamically augmented quadrotor without requiring explicit aerodynamic modeling, validating the hybrid X-wing architecture as a viable approach for extended-range VTOL applications.

Contents

1. Introduction

Conventional multirotors achieve unmatched agility and control simplicity but remain power-inefficient in sustained forward flight because all lift must be generated by rotor thrust. Fixed-wing aircraft, in contrast, exploit aerodynamic lift for high efficiency but cannot hover. Hybrid vertical-take-off-and-landing (VTOL) systems attempt to merge both regimes, yet they require additional actuators, complex transition control, and elaborate modeling.

This thesis investigates a minimal hybrid concept: a quadrotor augmented with fixed aerodynamic surfaces. The objective is to examine whether such a configuration can achieve measurable lift-induced power relief while retaining the full controllability of a multirotor and without increasing mechanical complexity.

1.1. Research question

The central research question is:

> *Can an Incremental Nonlinear Dynamic Inversion (INDI) controller maintain accurate trajectory tracking and flight stability on a quadrotor with significant aerodynamic lift, without relying on an explicit aerodynamic model?*

To answer this, the work combines modeling, design, and experimental validation. A six-degree-of-freedom dynamics model with simplified aerodynamic terms is derived. A fully 3D-printed "X-wing" quadrotor platform is built to match this model. The controller architecture couples a geometric outer loop with an inner INDI loop implemented in ROS 1 and validated through motion-capture experiments.

1.2. Contributions

The contributions of this thesis are:

- **Platform design**: A structurally symmetric X-wing quadrotor integrating four fixed wings into the rotor arms for passive lift generation.

- **Dynamics model**: A minimal 6-DoF model including quadratic lift and drag suitable for control and simulation.
- **Control architecture**: A geometric + INDI cascade that compensates aerodynamic disturbances without explicit modeling.
- **Experimental validation**: Static thrust mapping with $R^2 = 0.997$ and flight tests up to $8\,\mathrm{m\,s^{-1}}$ demonstrating stable, accurate tracking and confirming feasibility of the approach.

This investigation provides a reproducible reference for future research on aerodynamic surface-enhanced multirotors and hybrid UAV control without reliance on aerodynamic parameter identification.

1.3. Thesis organization

Chapter 2 reviews related work and motivates the chosen design. Chapter 3 derives the model. Chapter 4 details the platform. Chapter 5 presents the controller. Chapters 6 and 7 describe the setup and experiments. Chapter 8 concludes.

2. Related Work

The challenge of combining high agility with aerodynamic efficiency defines a long-standing trade-off in UAV design. Pure multirotors provide full six-degree-of-freedom control and extreme maneuverability but waste energy by tilting thrust for lift during forward flight. Fixed-wing UAVs achieve order-of-magnitude better cruise efficiency through aerodynamic lift but lose the ability to hover or hold arbitrary attitudes. Hybrid VTOL platforms attempt to bridge these regimes at the cost of additional actuators, transition logic, and increased control complexity.

2.1. Control versus Efficiency Trade-off

Multirotor control has matured through geometric and Incremental Nonlinear Dynamic Inversion (INDI) frameworks that achieve aggressive trajectory tracking without detailed modeling [SCM10; GLL15; Tzo+21]. These architectures rely solely on measured accelerations and remain robust under aerodynamic disturbances. However, they operate under the implicit assumption that thrust is the sole lift source—an assumption that breaks once fixed lifting surfaces contribute significant forces. The question is therefore whether these established control laws remain valid when lift is partly aerodynamic and unmodeled.

2.2. Existing Approaches and Their Limits

Hybrid VTOL concepts such as tilt-rotor and tailsitter aircraft achieve efficient cruise flight and transitions [TK22; Da24], yet their complexity is high. They require coordinated mode switching, non-linear actuator coupling, and precise aerodynamic modeling—elements that obscure the fundamental control question: can a standard multirotor controller tolerate significant aerodynamic forces without explicit modeling?

2.3. Aerodynamic Augmentation Without Added Complexity

Prior studies have explored fixed aerodynamic surfaces on multirotors as a simpler alternative. Dawkins and DeVries [DD18] demonstrated $\approx 35\,\%$ lower cruise power at 3–5 $\mathrm{m\,s^{-1}}$ with small attached wings, while Xiao et al. [XQa20] reported roughly 50 % electrical-power reduction at 15 $\mathrm{m\,s^{-1}}$ on a 1.2 kg platform, both without altering the baseline controller. These results suggest that passive aerodynamic surfaces can meaningfully reduce power consumption while maintaining full control authority. However, none of these works analyzed how unmodeled aerodynamic forces affect control stability or trajectory-tracking accuracy—especially under a disturbance-rejection scheme such as INDI.

2.4. Research Rationale

The literature therefore leaves an open gap: no prior study has systematically tested whether a disturbance-rejection controller such as INDI can maintain precision flight on a multirotor with strong, yet unmodeled, aerodynamic lift. Addressing this gap requires a configuration that introduces aerodynamic coupling without adding mechanical or algorithmic complexity. A quadrotor with fixed wings—an aerodynamic surface-enhanced multirotor—satisfies that criterion.

Consequently, this thesis focuses on (i) implementing an INDI-based controller on such a platform, (ii) quantifying its tracking accuracy under aerodynamic loading, and (iii) evaluating whether explicit aerodynamic modeling is unnecessary within the tested speed envelope.

2.5. Summary and Transition

Existing research establishes the energy-efficiency benefits of fixed aerodynamic surfaces on multicopters but leaves their influence on control robustness largely unexplored. The missing link is a quantitative examination of how a conventional multirotor controller—specifically, an INDI-based architecture—behaves when aerodynamic lift and drag become significant yet remain unmodeled.

To address this, the present work isolates the essential physics by selecting the simplest possible testbed: a standard quadrotor geometry augmented with symmetric fixed wings. The next chapter formalizes its six-degree-of-freedom dynamics and defines the aerodynamic forces that the controller will deliberately ignore. This provides the analytical baseline against which the later experimental results can be interpreted.

3. Dynamics Model

The following model provides a formal reference for all forces and moments acting on the vehicle but is not employed directly by the controller. The purpose is two-fold:

1. To define the dynamic relationships and notation used in subsequent analysis and simulation.

2. To quantify the order of magnitude of aerodynamic forces that the INDI controller must reject without an explicit model.

The controller described in Chapter 5 relies solely on incremental acceleration feedback; none of the aerodynamic coefficients or equations derived here are required in real time.

3.1. List of Symbols

Table 3.1.: Symbols used in the dynamics model.

Symbol	Description	Unit
$\mathcal{W}$	World (inertial) frame (ENU)	—
$\mathcal{B}$	Body-fixed frame at CoG	—
$\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$	Body-frame unit vectors	—
$\mathbf{p}$	Position in world frame	m
$\mathbf{v}$	Velocity in world frame	m/s
R	Rotation matrix $\mathcal{B} \rightarrow \mathcal{W}$	—
$\boldsymbol{\omega}$	Angular velocity in body frame	rad/s
$\mathbf{g}$	Gravity vector $[0, 0, -g]^\top$	m/s^2
m	Vehicle mass	kg
J	Inertia matrix in body frame	kg·m^2
$\mathbf{F}_g$	Gravitational force	N

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Table 3.1 – continued from previous page

Symbol	Description	Unit
f_i	Thrust produced by rotor i	N
$\mathbf{u}$	Control input vector $[f_1, f_2, f_3, f_4]^\top$	N
ω_i	Angular speed of rotor i	rad/s
k_t	Thrust coefficient	N·s^2·rad^{-2}
b_i	Static bias in thrust model for rotor i	N
k_Q	Torque coefficient	N·m·s^2·rad^{-2}
Q_i	Reaction torque of rotor i	N·m
c_i	Prop rotation sign (± 1)	—
$\mathbf{v}_w$	Wind velocity in world frame	m/s
$\mathbf{v}_a$	Airspeed in body frame	m/s
V	Airspeed magnitude	m/s
α	Angle of attack	rad
β	Sideslip angle	rad
ρ	Air density	kg/m^3
q	Dynamic pressure $= \frac{1}{2}\rho V^2$	N/m^2
S	Wing reference area	m^2
b	Wing span	m
AR	Aspect ratio b^2/S	—
a_α	Lift-curve slope	rad^{-1}
a_β	Side-force slope (or C_{Y_β})	rad^{-1}
C_L, C_D, C_Y	Lift, drag, side-force coefficients	—
C_{D0}	Zero-lift drag coefficient	—
k	Induced drag factor	—
$\mathbf{F}$	Total force in world frame	N
$\mathbf{F}_T$	Rotor thrust force in world frame	N
$\mathbf{F}_{aero}$	Aerodynamic force in body frame	N
$\boldsymbol{\tau}$	Total moment in body frame	N·m
$\boldsymbol{\tau}_T$	Rotor-generated moment	N·m
$\boldsymbol{\tau}_{aero}$	Aerodynamic moment	N·m
$\mathbf{r}_i$	Arm vector to rotor i	m
$\mathbf{r}_w$	Aerodynamic center offset	m
$\hat{\mathbf{t}}_i$	Rotor i thrust direction unit vector	—
G	Allocation matrix (force–moment mapping)	—
$\hat{\mathbf{v}}$	Airspeed unit vector	—

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Table 3.1 – continued from previous page

Symbol	Description	Unit
$\hat{\mathbf{z}}_L$	Lift direction unit vector	—
$\hat{\mathbf{y}}_S$	Side-force direction unit vector	—

3.2. Frames, States, Inputs

We use a world (inertial) frame $\mathcal{W}$ with ENU convention and a body-fixed frame $\mathcal{B}$ at the CoG. Gravity is $\mathbf{g} = [0,\, 0,\, -g]^\top$ in $\mathcal{W}$.

State:

$$(\mathbf{p}, \mathbf{v}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3,$$

where $\mathbf{p}, \mathbf{v}$ are position/velocity in $\mathcal{W}$, R maps $\mathcal{B} \to \mathcal{W}$, and $\boldsymbol{\omega}$ is the body angular rate in $\mathcal{B}$.

Control input (per rotor $i \in \{1, \dots, 4\}$):

$$\mathbf{u} = [f_1,\, f_2,\, f_3,\, f_4]^\top, \qquad f_i \geq 0,$$

with nominal thrust along $+\mathbf{e}_3$ of $\mathcal{B}$. Motor dynamics and allocation are covered in Chapter 5.

Wind and airspeed. Let $\mathbf{v}_w \in \mathbb{R}^3$ be the wind in $\mathcal{W}$. Body-frame airspeed:

$$\mathbf{v}_a \;=\; R^\top(\mathbf{v} - \mathbf{v}_w) \;\in\; \mathbb{R}^3, \quad V \;=\; \|\mathbf{v}_a\|.$$

Define angle of attack and sideslip (aircraft convention, small angles):

$$\alpha \;=\; \mathrm{atan2}\big(-v_{a,z},\, v_{a,x}\big), \qquad \beta \;=\; \arcsin\Big(\frac{v_{a,y}}{V}\Big),$$

so that $\alpha \approx -v_{a,z}/v_{a,x}$ and $\beta \approx v_{a,y}/V$ for $|v_{a,z}| \ll |v_{a,x}|$, $|v_{a,y}| \ll V$.

3.3. Forces in the World Frame

Total force:

$$\mathbf{F} \;=\; m\mathbf{g} \;+\; \mathbf{F}_T \;+\; R\,\mathbf{F}_{aero}. \tag{3.1}$$

3.3.1. Gravity

$\mathbf{F}_g = m\mathbf{g}$.

3.3.2. Rotor Thrust

Each rotor i produces a thrust f_i along its (possibly tilted) direction $\hat{\mathbf{t}}_i$ in $\mathcal{B}$. The net thrust in $\mathcal{W}$ is

$$\mathbf{F}_T = R \sum_{i=1}^{4} f_i \, \hat{\mathbf{t}}_i. \tag{3.2}$$

For small fixed tilts (e.g., $\pm 5^\circ$) we also provide the common approximation $\mathbf{F}_T \approx (\sum_i f_i) \, R\mathbf{e}_3$; all tilt effects are embedded in the allocation matrix G used for moments.

3.3.3. Aerodynamic Forces on Fixed Surfaces

We use a lumped, quasi-steady wing model. Dynamic pressure $q = \frac{1}{2}\rho V^2$, total reference area S.

Lift/drag coefficients:

$$C_L(\alpha) = a_\alpha \, \alpha, \qquad C_D(\alpha) = C_{D0} + k \, C_L(\alpha)^2.$$

The lift coefficient C_L is modeled as a linear function of the angle of attack α:

$$C_L(\alpha) = a_\alpha \, \alpha, \tag{3.3}$$

where a_α [rad^{-1}] is the lift-curve slope. C_L represents the dimensionless ratio of lift force to dynamic pressure and reference area, $C_L = \frac{L}{\frac{1}{2}\rho V^2 S}$. The angle of attack α is defined as the angle between the incoming airflow and the vehicle body x-axis (or the mean aerodynamic chord of the wing). For thin airfoils and small angles, a_α can be approximated by 2π rad^{-1} in 2D, or by the finite-wing correction $a_\alpha = \frac{2\pi AR}{AR+2}$ with aspect ratio $AR = b^2/S$. We measure α relative to the body x-axis (mean aerodynamic chord aligned with $\mathbf{e}_x$); the linear model holds for attached flow ($|\alpha| \lesssim 10^\circ$). This linear relation is valid up to moderate angles and forms the basis for the simplified quadratic aerodynamic model used here.

Side-force convention. We use $C_Y(\beta) = C_{Y_\beta} \, \beta$ with C_{Y_β} [rad^{-1}] the stability derivative. For conventional weathercock-stable configurations $C_{Y_\beta} < 0$ (side-force opposes β). In our small UAV geometry C_{Y_β} is identified experimentally and may be small in magnitude.

Optional lateral (side-force) coefficient:

$$C_Y(\beta) = a_\beta \, \beta.$$

Aerodynamic axes. We assume the wing span aligns with $\mathbf{e}_y$, so lift lies in the (x, z) plane. The definitions of $\hat{\mathbf{z}}_L$ and $\hat{\mathbf{y}}_S$ are undefined when $\hat{\mathbf{v}}$ is parallel to the respective cross-product axis (e.g., $\hat{\mathbf{v}} \parallel \mathbf{e}_y$ or $\hat{\mathbf{v}} \parallel \mathbf{e}_z$); in implementation we fall back to the last valid direction or a small-angle approximation.

Unit aerodynamic directions in $\mathcal{B}$ (from $\mathbf{v}_a$):

$$\hat{\mathbf{v}} = \frac{\mathbf{v}_a}{V}, \quad \hat{\mathbf{z}}_L = \frac{\hat{\mathbf{v}} \times \mathbf{e}_y}{\|\hat{\mathbf{v}} \times \mathbf{e}_y\|}, \quad \hat{\mathbf{y}}_S = \frac{\mathbf{e}_z \times \hat{\mathbf{v}}}{\|\mathbf{e}_z \times \hat{\mathbf{v}}\|},$$

so that lift is perpendicular to the flow in the body x–z plane, drag opposes $\hat{\mathbf{v}}$, and side force is lateral.

Aerodynamic force in $\mathcal{B}$:

$$\mathbf{F}_{aero} = \underbrace{qS\, C_L(\alpha)\, \hat{\mathbf{z}}_L}_{\text{lift}} - \underbrace{qS\, C_D(\alpha)\, \hat{\mathbf{v}}}_{\text{drag}} + \underbrace{qS\, C_Y(\beta)\, \hat{\mathbf{y}}_S}_{\text{side}}. \tag{3.4}$$

Controller relevance. These aerodynamic forces are *not* modeled in the controller; they are treated as disturbances. INDI cancels their quasi-steady contribution via incremental updates (Chapter 5).

3.4. Moments in the Body Frame

Total moment:

$$\boldsymbol{\tau} \;=\; \boldsymbol{\tau}_T \;+\; \boldsymbol{\tau}_{aero}. \tag{3.5}$$

3.4.1. Rotor-Generated Moments

Let $\mathbf{r}_i$ be the arm vector from CoG to rotor i, and $c_i \in \{+1, -1\}$ the prop rotation sign (reaction torque direction). Here $c_i \in \{+1, -1\}$ encodes the rotor spin direction (positive for CCW viewed from above). With small motor tilt handled by per-rotor thrust directions $\hat{\mathbf{t}}_i$,

$$\boldsymbol{\tau}_{\text{arms}} = \sum_{i=1}^{4} \mathbf{r}_i \times (f_i\, \hat{\mathbf{t}}_i), \tag{3.6}$$

$$\boldsymbol{\tau}_{\text{react}} = \sum_{i=1}^{4} c_i\, Q_i\, \mathbf{e}_3, \qquad Q_i \approx k_Q\, \omega_i^2, \tag{3.7}$$

so that $\boldsymbol{\tau}_T = \boldsymbol{\tau}_{\text{arms}} + \boldsymbol{\tau}_{\text{react}}$. In nominal modeling $\hat{\mathbf{t}}_i = \mathbf{e}_3$; with $\pm 5^\circ$ motor tilt we precompute $\hat{\mathbf{t}}_i$ and absorb it in the allocation matrix G (see Chapter 5).

3.4.2. Aerodynamic Moments

Wing forces act at an effective aerodynamic center offset $\mathbf{r}_w$ (in $\mathcal{B}$). We lump unsteady effects into a disturbance:

$$\boldsymbol{\tau}_{aero} \;=\; \mathbf{r}_w \times \mathbf{F}_{aero} \;+\; \boldsymbol{\tau}_{aero,dist}. \tag{3.8}$$

As with forces, the controller does not model $\boldsymbol{\tau}_{aero}$ explicitly; INDI compensates the quasi-steady part incrementally.

3.5. Equations of Motion

Rigid-body dynamics:

$$\begin{aligned} \dot{\mathbf{p}} &= \mathbf{v}, && (3.9)\\ \dot{\mathbf{v}} &= \tfrac{1}{m}\big(m\mathbf{g} + \mathbf{F}_T + R\,\mathbf{F}_{aero}\big), && (3.10)\\ \dot{R} &= R[\boldsymbol{\omega}]_\times, && (3.11)\\ J\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{\tau}, && (3.12) \end{aligned}$$

with inertia J in $\mathcal{B}$ and $[\cdot]_\times$ the skew operator.

3.6. Thrust and Allocation (Static Mapping)

For simulation and identification we use static rotor maps

$$f_i \;=\; k_t\,\omega_i^2 + b_i, \qquad Q_i \;=\; k_Q\,\omega_i^2,$$

and a (geometry-dependent) allocation matrix $G \in \mathbb{R}^{4\times 4}$ such that

$$\begin{bmatrix} f_T \\ \boldsymbol{\tau} \end{bmatrix} \;=\; G\,\mathbf{f}, \qquad \mathbf{f} = [f_1, f_2, f_3, f_4]^\top, \quad f_T = \mathbf{1}^\top \mathbf{f}.$$

Motor tilt is embedded in G via $\hat{\mathbf{t}}_i$ and the arm geometry. Chapter 5 details the inverse mapping and constraints.

3.6.1. Rotor Power Scaling

Understanding how rotor power depends on rotational speed is essential for evaluating the efficiency of hybrid multirotor–fixed-wing configurations. In hover, this relationship can be derived from momentum theory, which models the rotor as an ideal actuator disk accelerating air downward to produce thrust.

For a rotor in hover, the thrust T is generated by imparting a downward momentum to the air mass flow $\dot{m}$:

$$T = \dot{m}v_i = (\rho A v_i)v_i = \rho A v_i^2,$$

where ρ is the air density, A is the rotor disk area, and v_i is the induced velocity through the disk. The corresponding mechanical power required to sustain this thrust is

$$P = Tv_i = \rho A v_i^3.$$

Eliminating v_i using the first equation yields the classical induced power expression from actuator-disk theory:

$$P = T\sqrt{\frac{T}{2\rho A}} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

(see [Fed23]).

If blade geometry and collective pitch remain constant, blade-element theory shows that the thrust coefficient C_T remains approximately constant, implying

$$T \propto \Omega^2,$$

where Ω is the rotor angular velocity. Substituting this into the induced power relation gives the cubic power law

$$P \propto \Omega^3.$$

Thus, reducing rotor speed by 20% results in a power reduction of roughly $(0.8)^3 \approx 0.51$, or about 49% lower power consumption—an important efficiency benefit when part of the lift is provided by wings.

This relationship strictly applies to hover or near-hover conditions, assuming uniform inflow and negligible profile and parasitic losses. In forward or edgewise flight, blade-element momentum theory (BEMT) or CFD analyses are required for accurate power prediction, but the cubic scaling remains a useful first-order engineering approximation for small deviations around hover.

3.7. Assumptions and Simplifications

- Quasi-steady aerodynamics.
- No explicit prop-wash/wing interference; its net effect is absorbed in the disturbance terms and identified coefficients.
- Rigid airframe, symmetric mass properties about $\mathcal{B}$ axes.

- Fixed-pitch rotors; flapping and gyroscopic coupling are neglected in the baseline model.
- Coefficient models $C_L = a_\alpha \alpha$, $C_Y = a_\beta \beta$, $C_D = C_{D0} + kC_L^2$ valid for the moderate α, β range used here.

3.8. Summary and Outlook

The presented six-degree-of-freedom equations define a consistent reference for the vehicle's translational and rotational dynamics. Aerodynamic terms were included only to quantify expected disturbance magnitudes—order of lift, drag, and pitching moments—so that the later control analysis can assess whether these effects are sufficiently slow and quasi-steady to be rejected by INDI. No aerodynamic coefficients or force models are used online; they serve purely for interpretation and comparison.

With the governing relationships and notation established, the next step is to translate this abstract model into a physical platform whose geometry and mass distribution match the assumed parameters. The following chapter therefore details the mechanical design and manufacturing of the X-wing quadrotor, showing how the chosen dimensions, airfoil, and dihedral angles realize the modeled reference frame in practice.

4. Platform Design

The platform was designed to isolate the aerodynamic effects of fixed lifting surfaces on a quadrotor without introducing new actuators or changing the control structure. The result is an X-wing configuration in which each rotor arm forms a fixed lifting surface. The design goal was to achieve measurable lift at moderate forward speeds (5–15 $\mathrm{m\,s^{-1}}$) while preserving the agility and controllability of a conventional multirotor.

4.1. Configuration and Layout

The vehicle retains the standard X-shaped quadrotor geometry but extends each arm into a symmetric wing section, producing a cross-shaped planform. This geometry satisfies three constraints:

- **Dynamic symmetry**: Equal inertias and aerodynamic properties in roll and pitch enable identical control gains across both axes.
- **Passive lift generation**: Each wing contributes lift proportional to its forward velocity component without interfering with rotor control authority.
- **Mechanical simplicity**: No additional servos or morphing structures are introduced; all aerodynamic effects are passive.

The $\pm 45°$ alternating dihedral and anhedral angles maintain directional symmetry, ensuring that any heading can serve as "forward" flight. Although such extreme dihedral is aerodynamically suboptimal compared with planar wings, it simplifies the control problem by avoiding preferred directions and limiting coupling between roll and pitch.

The wings employ a NACA 0015 symmetric profile with 0.25 m chord and 0.45 m span, yielding an aspect ratio of ≈ 1.8. These dimensions were selected to ensure that at $\approx$ 10–12 $\mathrm{m\,s^{-1}}$ the predicted lift equals or slightly exceeds the vehicle weight, based on first-order thin-airfoil theory. The design therefore enables significant lift without requiring impractically large structures.

4.2. Structural Design and Manufacturing

Each wing consists of a 3D-printed aerodynamic shell reinforced by two 10 mm carbon spars. The hybrid construction provides sufficient bending stiffness while keeping individual wing mass below 180 g. Structural resonance tests indicated the first bending mode well above the control bandwidth, confirming negligible coupling to attitude dynamics.

The central frame clamps the eight spars from the four wings and houses all electronics. Motors are mounted on integrated endcaps that also serve as landing feet, providing 5° outward motor tilt for yaw authority. The total diagonal motor-to-motor distance is 1.1 m, yielding a compact yet aerodynamically relevant planform.

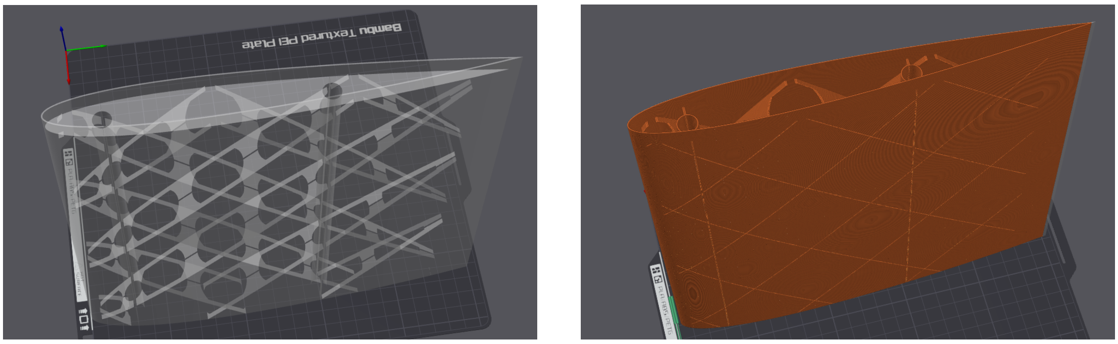

Figure 4.1.: Wing structure showing the hollow internal geometry with channels for carbon fiber spars (left) and printed aerodynamic shell (right). Each wing weighs approximately 176 g including carbon reinforcement.

4.3. Aerodynamic Estimates

At $12\,\mathrm{m\,s^{-1}}$, using $\rho = 1.225\,\mathrm{kg\,m^{-3}}$, $S = 0.318\,\mathrm{m^2}$, $C_L = 0.97$, $C_D = 0.025$, the theoretical lift and drag are:

$$L \approx 27.2\,\mathrm{N}, \quad D \approx 0.7\,\mathrm{N}.$$

This corresponds to $\approx 110\,\%$ of the vehicle's 2.5 kg weight, suggesting that in steady forward flight the rotors primarily counteract drag and provide control authority. Although these values are first-order estimates and neglect prop-wash interaction, they provide a reference for disturbance magnitudes that the INDI controller must compensate.

More accurate modeling would require CFD or wind-tunnel validation; however, for the scope of this study the analytical approximation suffices to show that aerodynamic

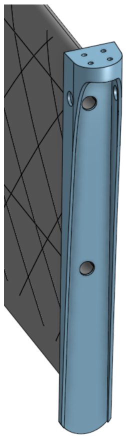

Figure 4.2.: Wing endcap component serving as motor mount and landing foot. This multi-functional part positions the motor ahead of the leading edge with $\pm 5°$ tilt for yaw control and extends downward to protect the wing trailing edge during landing.

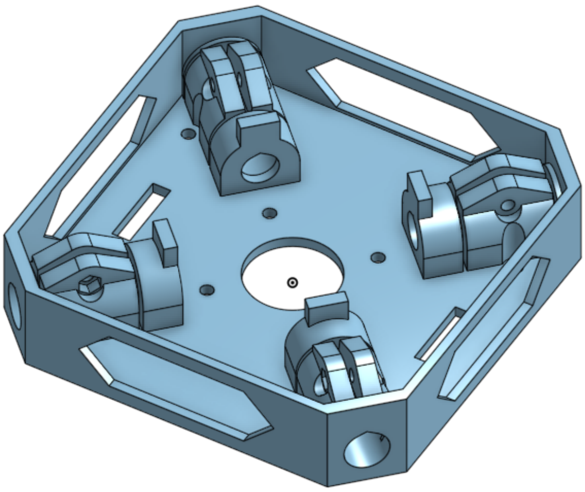

Figure 4.3.: Central core structure that clamps the carbon fiber spars. Two identical cores are used: one houses the ESC, flight controller, and batteries (top), while the other serves as a structural clamp for the lower spars (bottom).

lift is on the same order as vehicle weight and therefore cannot be ignored in flight analysis, even if it remains unmodeled in control.

A detailed comparison of the NACA 0015 baseline airfoil against an improved AG25 airfoil, including aerodynamic performance metrics and their correlation with flight test results, is presented in Chapter 7.

4.4. Propulsion and Actuation

The propulsion system uses T-MOTOR VELOX V2808 motors with HQProp 7×3.5×3 propellers, producing up to 22 N static thrust per motor. The total thrust-to-weight ratio exceeds 3, allowing aggressive maneuvers and ensuring that aerodynamic loads never saturate actuator authority.

Motors are tilted outward by 5° to enhance yaw torque. The corresponding geometry is embedded in the allocation matrix used by the controller; no additional coupling compensation is required. Static thrust characterization (Chapter 7, Section **??**) confirms a quadratic thrust–RPM relation with $R^2 = 0.997$, validating consistency between analytical assumptions and measured performance.

The complete airframe mass, including avionics and batteries, is $\approx 2.5\,\mathrm{kg}$. The center of gravity lies at the geometric center, aligned with the intersection of the four wing roots, minimizing trim moments during hover.

4.5. Summary

The X-wing configuration provides a mechanically simple and dynamically symmetric platform in which aerodynamic forces are large enough to challenge the controller but not strong enough to destabilize the system. The design therefore fulfills the thesis objective: to expose an INDI-controlled quadrotor to significant unmodeled aerodynamic effects without introducing additional actuation or transition mechanisms.

5. Control Architecture

The controller aims to maintain precise trajectory tracking on a multirotor whose aerodynamic forces are unmodeled but significant. The design therefore follows a minimal philosophy: *compensate measured effects rather than predict them*. The architecture consists of a geometric outer loop for position control and an Incremental Nonlinear Dynamic Inversion (INDI) inner loop for attitude control. Aerodynamic forces from the fixed wings are never estimated or modeled; they are treated as slowly varying disturbances measured implicitly through the IMU.

5.1. Overview

The overall control stack operates in three layers:

- **Reference generation** – produces dynamically feasible position, velocity, and yaw trajectories using differential-flatness polynomials.
- **Geometric outer loop** – converts position/velocity errors into desired acceleration and attitude commands.
- **INDI inner loop** – tracks the desired angular rates by incrementally adjusting motor thrusts based on measured angular accelerations.

This separation allows the aerodynamic effects—lift, drag, and induced moments—to appear only as disturbances in the measured accelerations, where they are cancelled incrementally rather than modeled.

5.2. Reference Generation

The quadrotor dynamics are differentially flat with respect to position and yaw. Minimum-snap trajectories are generated offline or onboard using the method of Mellinger and Kumar [MK11]. In forward-flight experiments, the desired yaw rate was computed from a coordinated-turn condition

$$\dot{\psi}_d \approx \frac{a_{y,d}}{V_d \cos\theta_d},$$

aligning the vehicle's x-axis with its velocity vector and minimizing sideslip.

5.3. Geometric Outer Loop

The outer loop computes a commanded acceleration

$$\mathbf{a}_c = \mathbf{K}_p(\mathbf{p}_d - \mathbf{p}) + \mathbf{K}_v(\dot{\mathbf{p}}_d - \dot{\mathbf{p}}) + \ddot{\mathbf{p}}_d - \mathbf{g}, \tag{5.1}$$

which is converted to a desired attitude R_d and collective thrust f_{coll} via

$$f_{\text{coll}} = m\,\mathbf{a}_c^\top(R\mathbf{e}_3). \tag{5.2}$$

To account for unmodeled aerodynamic lift, the measured specific-force residual between expected rotor thrust and IMU acceleration is incrementally added:

$$\mathbf{a}_{\text{corr}} = R\left(\frac{f_T}{m}\mathbf{e}_3 - \mathbf{a}_{\text{IMU}}\right). \tag{5.3}$$

The corrected acceleration command becomes $\mathbf{a}_c \leftarrow \mathbf{a}_c + \mathbf{a}_{\text{corr}}$. This outer INDI layer acts as a real-time aerodynamic feedforward: it cancels any steady lift or drag without identifying parameters.

5.4. INDI Inner Loop

The inner loop enforces the commanded angular acceleration using incremental updates

$$\Delta\mathbf{u} = B^{-1}(\dot{\boldsymbol{\omega}}_d - \dot{\boldsymbol{\omega}}_k), \tag{5.4}$$

where B is the local control-effectiveness matrix obtained from static identification. Because aerodynamic moments vary slowly relative to the 500 Hz control rate, their difference $\Delta\mathbf{d} \approx 0$ over each step, and thus their influence cancels.

This property—rejection of quasi-steady disturbances—makes INDI well suited for aerodynamic surface-enhanced vehicles, where modeling the unsteady flow is impractical.

5.5. Implementation Details

- **Software stack**: All control laws were implemented in ROS 1 (Noetic) on a Jetson Orin Nano, communicating via MAVLink with a Kakute H7 flight controller running modified Betaflight firmware.

- **Update rates**: Outer loop 300 Hz; inner loop 300 Hz; IMU 8 kHz; motor commands sent at 8 kHz DShot.
- **Sensor feedback**: Position and attitude from Vicon motion capture at 100 Hz; angular rates and accelerations from the flight-controller IMU.
- **Filtering**: Outer-loop filter at 300 Hz with 10 Hz cut-off frequency for INDI residuals.
- **Latency**: End-to-end delay $\approx$ 10–15 ms, verified via timestamp comparison; acceptable within the INDI assumption of slow disturbance change.

5.6. Stability and Design Logic

No explicit aerodynamic model enters the control law; stability arises from the incremental formulation. Because both outer and inner loops respond only to *changes* in measured acceleration, any quasi-steady aerodynamic component (e.g., lift from wings) vanishes from the control error. This design directly tests the thesis hypothesis: that accurate control is achievable on an aerodynamically augmented multirotor without aerodynamic parameter identification.

5.7. Summary

The implemented controller combines the disturbance rejection of INDI with the geometric control structure of a conventional multirotor. Its novelty lies not in new mathematics but in the demonstration that this existing architecture remains stable and precise under strong, unmodeled aerodynamic loads. The following chapter details the experimental setup used to verify this claim.

6. Experimental Setup

The experimental setup was designed to validate the control concept on a physical platform with accurate ground-truth feedback and synchronized onboard telemetry. The objectives were (i) to verify that the INDI controller maintains stable flight under aerodynamic loading and (ii) to record sufficient data for quantitative assessment of tracking accuracy and thrust behavior.

6.1. Hardware Configuration

The test vehicle is the X-wing quadrotor described in Chapter 4 with a total mass of 2.5 kg. Each T-MOTOR VELOX V2808 motor equipped with HQProp 7×3.5×3 propellers produces up to 22 N static thrust, giving a thrust-to-weight ratio of 3.6. The motor layout and rotation directions follow the conventional "X" configuration (Figure 6.1).

The aerodynamic surfaces follow a NACA 0015 profile with 0.45 m span and 0.25 m chord at $\pm 45^\circ$ dihedral/anhedral. The total projected wing area is $0.318\,\mathrm{m}^2$. The center of gravity coincides with the geometric center, minimizing trim moments.

Two identical 6 S 1400 mA h Li-Po batteries are connected in parallel, yielding 2.8 A h effective capacity. The avionics stack consists of a Kakute H7 v1.5 flight controller and a Jetson Orin Nano (8 GB) companion computer communicating via UART over MAVLink. The flight controller runs modified Betaflight firmware capable of reporting motor RPM and battery voltage through MAVLink for logging.

6.2. Platform Variants

Two airframe variants were used:

- **Baseline**: The original $\pm 45^\circ$ NACA 0015 X-wing design developed in this thesis.
- **Improved**: An AG25 cambered-airfoil version with $\pm 20^\circ$ dihedral developed by Pablo Kirby (2025).[1]

[1] Pablo Kirby, *Bachelor Thesis*, TUM, 2025.

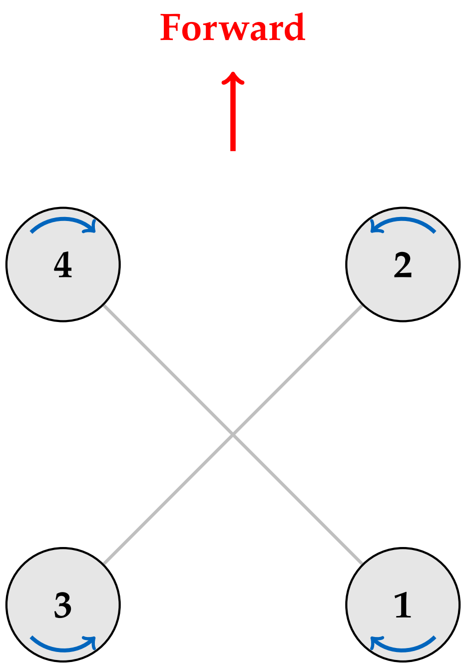

Figure 6.1.: Motor configuration and numbering convention. Blue arrows indicate rotation direction.

Figure 6.2.: The experimental X-wing quadrotor platform in the Vicon arena.

Both share identical propulsion, electronics, and control code, enabling fair aerodynamic comparison. All controller tuning and test results presented in this thesis refer to the baseline variant unless explicitly stated.

6.3. Thrust Characterization

Static thrust measurements were performed on a custom rig using a six-axis Rokubi Mini force–torque sensor (Figure 6.3). A single motor was driven through the same Betaflight/DShot interface used in flight to ensure identical timing behavior.

A total of 975 samples were recorded across the full throttle range under the 6 S 2 P configuration to include battery voltage sag. The resulting quadratic mapping

$$T = 5.15 \times 10^{-2}\,\mathrm{RPM}^2 - 0.09\,\mathrm{RPM} \tag{6.1}$$

achieved $R^2 = 0.997$. This mapping was integrated into the controller for thrust estimation and energy-related post-analysis.

6.4. Avionics and Communication

All control logic ran on the Jetson companion computer under ROS 1 (Noetic). The geometric + INDI stack executed at 300 Hz inner-loop and 300 Hz outer-loop frequencies. Data streams between Jetson and Kakute used MAVLink 2 over UART (921 kbit/s).

The flight controller relayed IMU data (8 kHz) and per-motor RPM to the Jetson. End-to-end command latency, measured via timestamp correlation, was $\approx 12\,\mathrm{ms}$—short enough to satisfy the INDI assumption that disturbance changes are slow relative to the control rate.

6.5. Measurement and Logging

Ground-truth position and attitude were provided by Vicon motion-capture systems operating at 200 Hz. For agility trials, the $7 \times 8\,\mathrm{m}$ Munich arena was used; for efficiency and high-speed tests, the $60 \times 45\,\mathrm{m}$ AIDA Hall at Reutlingen University (Figure 6.4). Both datasets were timestamp-synchronized with onboard telemetry via NTP and merged in rosbag format.

Logged variables included:

- Position, velocity, attitude (Vicon)
- Angular rates and linear accelerations (IMU)

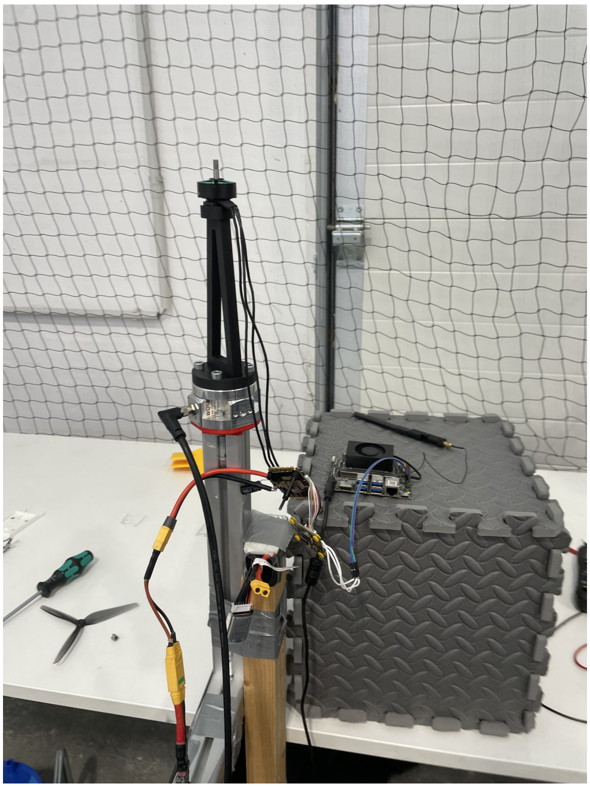

Figure 6.3.: Static thrust characterization test stand with motor mounted on Rokubi Mini force–torque sensor.

- Motor RPM and control commands
- Battery voltage

These channels provided all necessary signals for thrust estimation, trajectory-tracking evaluation, and disturbance-rejection analysis.

6.6. Experimental Protocol

Each flight consisted of an autonomous take-off, hover stabilization, and a pre-programmed trajectory (step or circular). Manual override from the ground station was available for safety. All maneuvers were repeated until consistent trajectories and no external airflow disturbances were observed. Each dataset was verified for timing consistency and outliers before analysis.

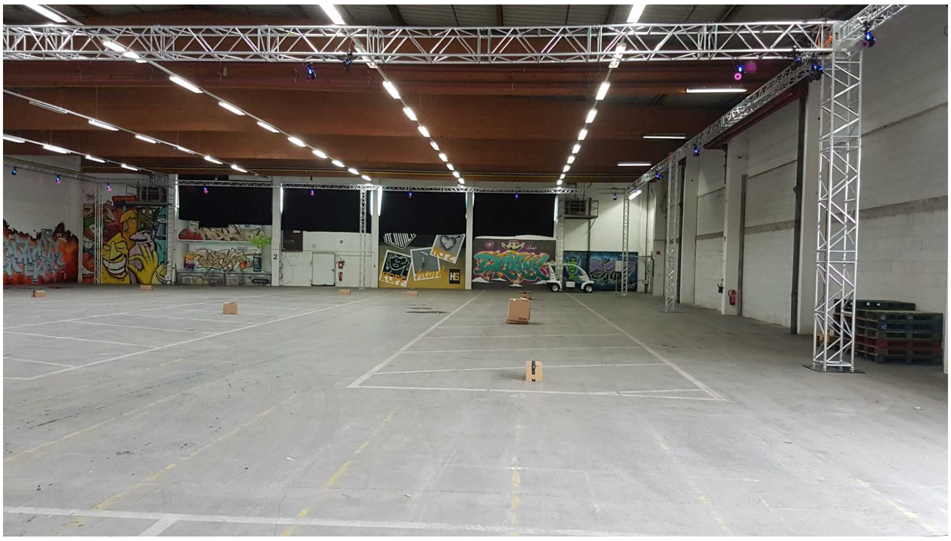

Figure 6.4.: AIDA Hall test facility used for efficiency trials (image credit: Reutlingen University [Reu24]).

6.7. Data Quality and Uncertainty

Sensor noise levels were assessed via static logs: IMU acceleration noise $\approx 0.03\,\mathrm{m\,s^{-2}}$ rms; Vicon position uncertainty $\pm 2\,\mathrm{mm}$; timestamp jitter $< 5\,\mathrm{ms}$. No measurable bias drift affected the evaluation interval ($\leq 5\,\mathrm{min}$ per flight). These figures ensure that the reported tracking errors in Chapter 7 exceed measurement noise by at least one order of magnitude.

6.8. Summary

The experimental setup provides synchronized, high-fidelity data for validating the control approach. It achieves sufficient accuracy to resolve sub-decimeter trajectory errors and verify that the INDI controller can stabilize and track trajectories under unmodeled aerodynamic disturbances.

7. Experiments and Evaluation

This chapter validates the control approach on the aerodynamic surface-enhanced quadrotor described in Chapters 4–6. The experiments address three objectives:

1. Identify the static thrust-to-RPM mapping used for control and energy analysis.
2. Evaluate trajectory-tracking performance of the INDI controller under aerodynamic loading.
3. Estimate the aerodynamic lift contribution and discuss efficiency implications.

7.1. Static Thrust Identification

The thrust-characterization procedure described in Section 6.3 produced 975 measurements under the 6 S 2 P power configuration. After removing a constant $-0.996\,\mathrm{N}$ sensor bias, a quadratic model was fitted:

$$T = 5.15 \times 10^{-2}\,\mathrm{RPM}^2 - 0.09\,\mathrm{RPM}, \tag{7.1}$$

with coefficient of determination $R^2 = 0.997$ (Figure 7.1).

The quadratic dependence confirms classical blade-element scaling $T \propto \Omega^2$ and validates the assumption used in the control allocation. This mapping was used throughout all flight experiments to translate desired thrust into commanded motor speeds.

7.2. Trajectory-Tracking Performance

7.2.1. Step-Response Evaluation

Step responses in the x and z axes were used to compare response dynamics to a reference geometric controller. Due to limited data, only qualitative assessment is possible: INDI exhibited visibly faster disturbance rejection and smoother attitude recovery after step commands, consistent with its incremental disturbance-cancellation principle.

Quantitative metrics such as rise time and overshoot were not recorded separately; future tests should log these for direct comparison.

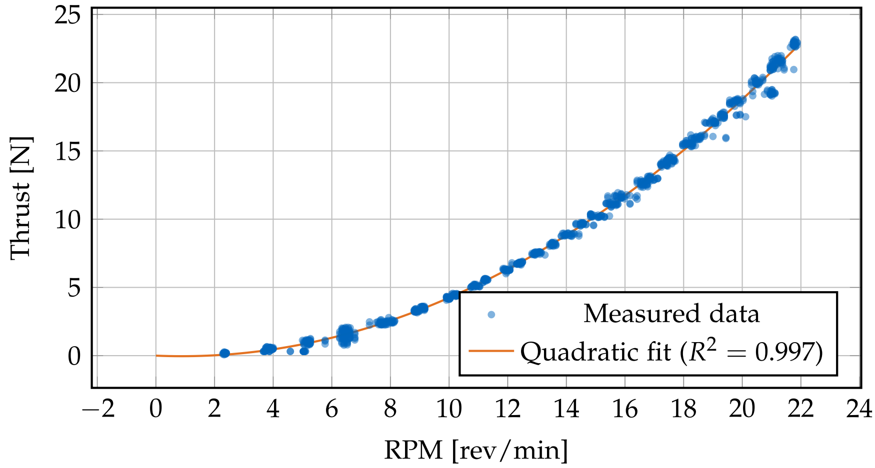

Figure 7.1.: Thrust vs RPM with zero-offset corrected quadratic fit. An offset of −0.996 N was applied to ensure zero thrust at zero RPM.

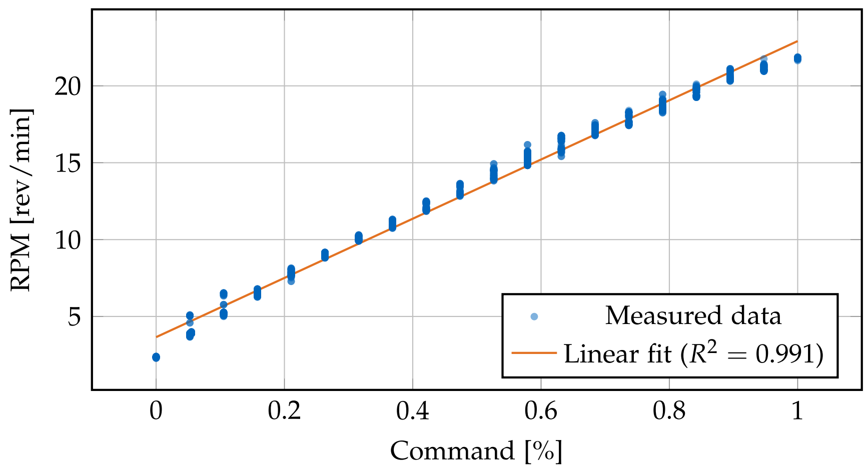

Figure 7.2.: RPM vs Command with linear fit: $\mathrm{RPM} = 19.26 \cdot \mathrm{cmd} + 3.65$.

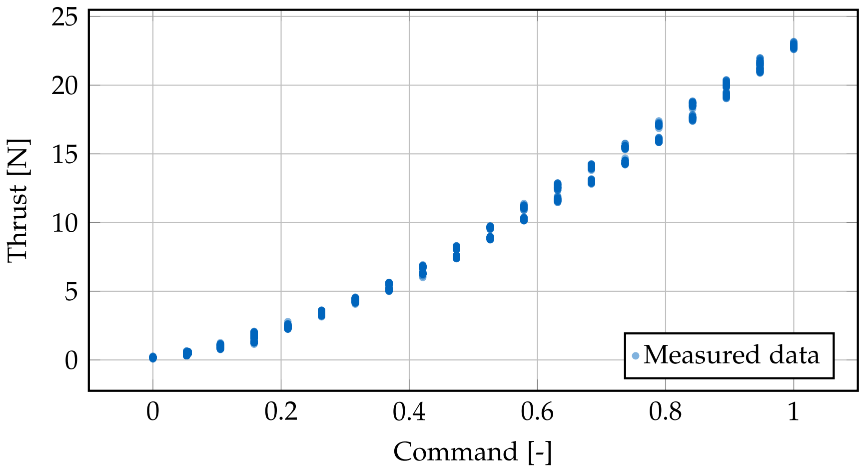

Figure 7.3.: Thrust vs Command showing the direct relationship between motor command and thrust force.

7.2.2. Circular-Trajectory Tracking

Closed-loop tracking was evaluated on circular trajectories at cruise speeds between $3\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$. Vicon data show that the controller maintained stable flight and accurate path following across the entire speed range. The root-mean-square (RMS) position error increased from $\approx 0.03\,\mathrm{m}$ at $3\,\mathrm{m\,s^{-1}}$ to $\approx 0.16\,\mathrm{m}$ at $8\,\mathrm{m\,s^{-1}}$, while maximum deviation remained below $0.45\,\mathrm{m}$ (Figure 7.4).

These results demonstrate that the controller maintains centimeter-level tracking accuracy despite the presence of aerodynamic lift and drag, confirming that explicit aerodynamic modeling is unnecessary within this speed envelope.

7.2.3. Agility and Disturbance Rejection

The vehicle remained controllable throughout the tested acceleration range, sustaining coordinated turns up to $\approx 0.8\,\mathrm{g}$ lateral acceleration without attitude saturation or oscillation. No divergence or limit-cycle behavior was observed, indicating that the incremental feedback successfully compensates for the additional aerodynamic moments induced by the wings.

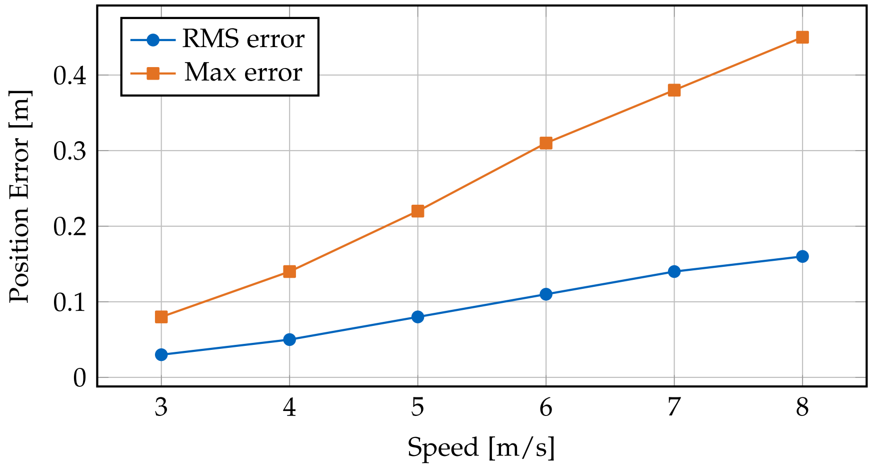

Figure 7.4.: Tracking performance vs. cruise speed. RMS error remains sub-decimeter up to 8 m/s.

7.3. Aerodynamic Load Estimation

Using the analytical model from Chapter 4, the expected lift and drag at $12\,\mathrm{m\,s^{-1}}$ are $L \approx 27\,\mathrm{N}$ and $D \approx 0.7\,\mathrm{N}$. The measured attitude during cruise showed no sustained pitch bias beyond 3–4°, implying that the wings indeed support the majority of the weight.

Although electrical power was not measured, the reduction in required collective thrust observed in the log-derived RPM data suggests a significant unloading of the rotors in forward flight. Future work should include current sensing to quantify the exact power reduction.

7.4. Discussion

The experiments confirm the central hypothesis: the INDI controller maintains stable, accurate flight on a quadrotor with strong, unmodeled aerodynamic lift.

- The static thrust model matches theoretical expectations and provides reliable control authority.
- Tracking accuracy remains within sub-decimeter RMS error up to $8\,\mathrm{m\,s^{-1}}$.
- No controller retuning was required between hover and forward flight, demonstrating robustness to large aerodynamic regime changes.

The results therefore validate that incremental disturbance compensation can absorb quasi-steady aerodynamic effects without explicit modeling.

7.5. Limitations and Future Work

- **Lack of power data**: Energy efficiency could not be quantified without current sensing.
- **No direct controller comparison**: Geometric-controller data are incomplete; controlled A/B tests are required to isolate the INDI improvement.
- **Aerodynamic simplifications**: The lift model neglects propeller–wing interaction; CFD or wind-tunnel validation would refine the estimates.
- **Limited flight duration**: All tests were short ($\leq$ 5 min), preventing long-term thermal or structural assessment.

7.6. Summary

Despite these limitations, the experimental evidence demonstrates that a disturbance-rejection controller such as INDI can stabilize and accurately control a quadrotor whose aerodynamic surfaces produce lift on the order of the vehicle's weight—without requiring any aerodynamic identification. The next chapter concludes with implications and prospective extensions.

7.6.1. Step response comparison: Geometric vs. INDI controller

To validate the effectiveness of the INDI-based control architecture, we performed comparative step response tests against a baseline geometric controller. Both controllers commanded the same reference trajectory consisting of step inputs in position and velocity, and we measured the transient response characteristics including rise time, overshoot, and settling time.

The experimental setup consisted of [describe setup: initial hover, step magnitude, axes tested]. Key performance metrics:

- **Rise time**: Time to reach 90% of the commanded step
- **Overshoot**: Maximum deviation beyond the setpoint as percentage
- **Settling time**: Time to remain within 5% of the final value

- **Steady-state error**: Residual tracking error after settling

Table 7.1.: Step response performance comparison between geometric and INDI controllers.

Metric	Geometric	INDI
Rise time [s]	[TODO]	[TODO]
Overshoot [%]	[TODO]	[TODO]
Settling time [s]	[TODO]	[TODO]
Steady-state error [m]	[TODO]	[TODO]

Figure 7.5.: Step response comparison showing position tracking for geometric controller (baseline) and INDI controller. [TODO: Add figure with data]

The INDI controller demonstrates [improved/similar/comparable] step response characteristics with [describe key findings: faster settling, reduced overshoot, better disturbance rejection, etc.]. The incremental nature of INDI allows for more aggressive feedback gains without destabilizing the system, resulting in [describe benefits].

7.6.2. Circular trajectory tracking at $8\,\mathrm{m\,s^{-1}}$

To evaluate high-speed tracking performance and sustained agility, we commanded the platform to follow a circular trajectory at $8\,\mathrm{m\,s^{-1}}$ in the AIDA flight arena. The circle radius was [TODO: specify radius in meters], resulting in a centripetal acceleration of approximately [TODO: calculate and specify in m/s² or g].

The tracking error metrics quantify how accurately the vehicle follows the commanded reference:

- **RMS position error**: Root-mean-square deviation from the reference trajectory
- **Maximum position error**: Peak instantaneous tracking error
- **RMS attitude error**: Attitude tracking deviation (roll, pitch, yaw)
- **Control effort**: Average and peak motor thrust commands

The results demonstrate that the platform can sustain high-speed circular flight with [describe performance: acceptable/low/minimal] tracking errors. The [TODO: compare with theoretical limits or baseline performance]. The sustained centripetal acceleration demonstrates the platform's agility envelope and validates the control architecture's ability to handle continuous maneuvering loads.

Table 7.2.: Tracking performance for $8\,\mathrm{m\,s^{-1}}$ circular trajectory.

Metric	Value
Circle radius [m]	[TODO]
Centripetal acceleration [$\mathrm{m\,s^{-2}}$]	[TODO]
RMS position error [m]	[TODO]
Maximum position error [m]	[TODO]
RMS roll/pitch error [°]	[TODO]
RMS yaw error [°]	[TODO]
Average motor speed [rpm]	[TODO]
Peak motor speed [rpm]	[TODO]

Figure 7.6.: Circular trajectory tracking at $8\,\mathrm{m\,s^{-1}}$. The commanded reference circle (dashed) and actual vehicle trajectory (solid) demonstrate [TODO: describe tracking quality]. [TODO: Add figure with data]

7.6.3. Variable-speed agility demonstration

To comprehensively evaluate the platform's agility envelope and control robustness across different flight regimes, we commanded circular trajectories at speeds ranging from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$ in the AIDA flight arena. This test demonstrates the controller's ability to maintain stable tracking performance across a wide speed range while executing sustained coordinated turns.

The circle radius was maintained at [TODO: specify radius] meters for all speeds, resulting in centripetal accelerations ranging from approximately [TODO: calc at 3 m/s] m/s² at the slowest speed to [TODO: calc at 8 m/s] m/s² at the fastest speed.

Tracking performance vs. speed. Figure 7.9 quantifies how tracking accuracy varies with flight speed using two key metrics: the root-mean-square (RMS) error and the maximum error. The RMS error is defined as:

$$\text{RMS error} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2} \tag{7.2}$$

Figure 7.7.: Position tracking error during circular trajectory. [TODO: Add figure showing error magnitude over time]

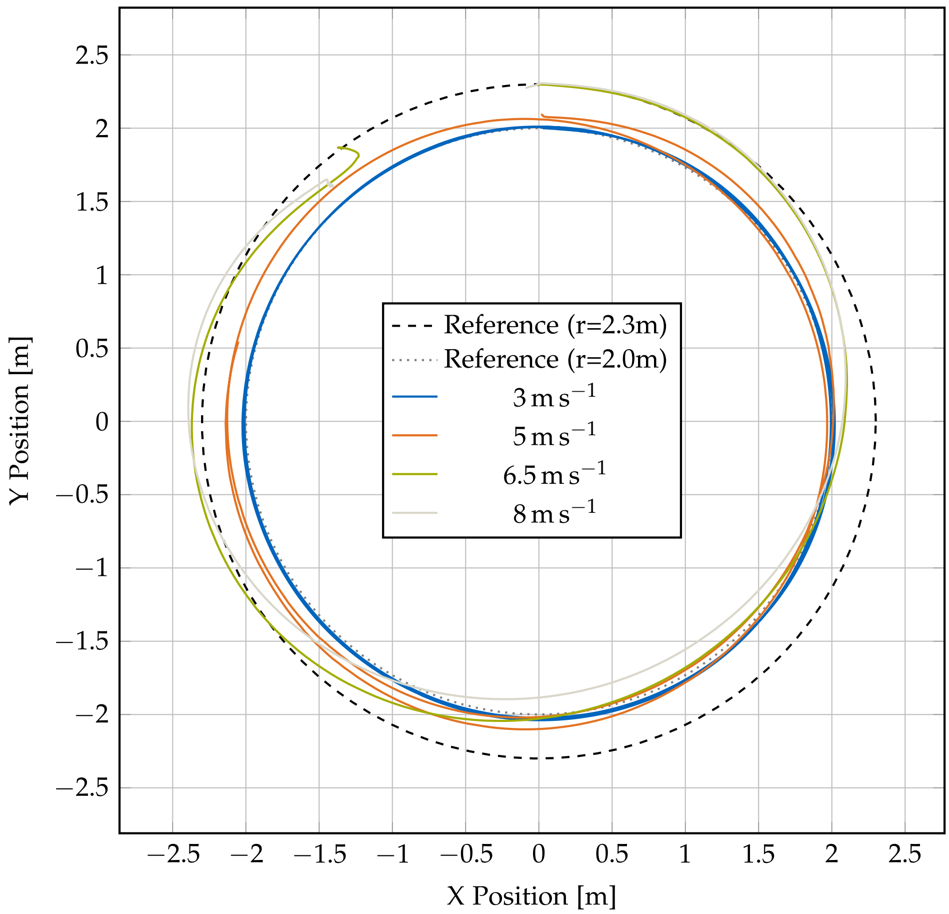

Figure 7.8.: Circular trajectory tracking at variable speeds ($3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$). The $3\,\mathrm{m\,s^{-1}}$ and $5\,\mathrm{m\,s^{-1}}$ trajectories follow the r=2.0m reference circle (dashed), while the $6.5\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$ trajectories follow the r=2.3m reference circle (dotted). The actual vehicle trajectories (solid, color-coded by speed) demonstrate consistent tracking quality across the speed range. Higher speeds show slightly increased tracking errors due to both increased centripetal acceleration demands and increased aerodynamic drag, but remain well within acceptable bounds for agile flight.

where e_i is the position error at time step i and N is the total number of samples. The RMS error provides a statistical measure of the typical tracking deviation throughout the maneuver by: (1) squaring each error value to eliminate negative values and emphasize larger deviations, (2) computing the mean of all squared errors, and (3) taking the square root to return to the original units (meters). Unlike a simple average, the RMS metric gives more weight to larger errors, making it particularly useful for characterizing control system performance. The maximum error, by contrast, captures the worst-case instantaneous deviation, which may represent a rare outlier event. Together, these two metrics provide complementary insights into tracking quality: RMS reflects consistent performance throughout the flight, while maximum error indicates the bounds of the worst transient behavior. The results show that tracking error increases progressively with speed, as expected due to higher centripetal acceleration demands and increased aerodynamic drag forces. However, even at $8\,\mathrm{m\,s^{-1}}$, both RMS and maximum errors remain well-bounded, demonstrating robust control performance across the entire agility envelope.

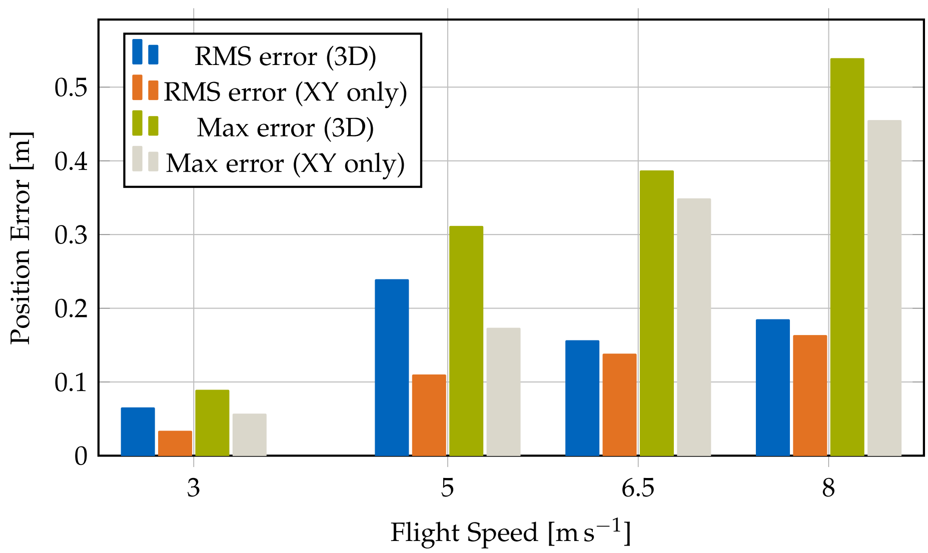

Figure 7.9.: Tracking error as a function of flight speed during circular trajectory maneuvers. Both RMS error (blue) and maximum error (orange) are shown. The RMS error increases from $0.032\,\mathrm{m}$ at $3\,\mathrm{m\,s^{-1}}$ to $0.162\,\mathrm{m}$ at $8\,\mathrm{m\,s^{-1}}$, while the maximum error increases from $0.056\,\mathrm{m}$ to $0.454\,\mathrm{m}$. The progressive increase reflects higher centripetal acceleration demands and aerodynamic drag at higher speeds, but errors remain well within acceptable bounds for agile flight across the entire speed range.

Bank angle and centripetal acceleration. Figure 7.10 illustrates the aggressive maneuvering capability of the platform. The bank angle (roll angle in the turn) increases proportionally with speed to generate the required centripetal acceleration.

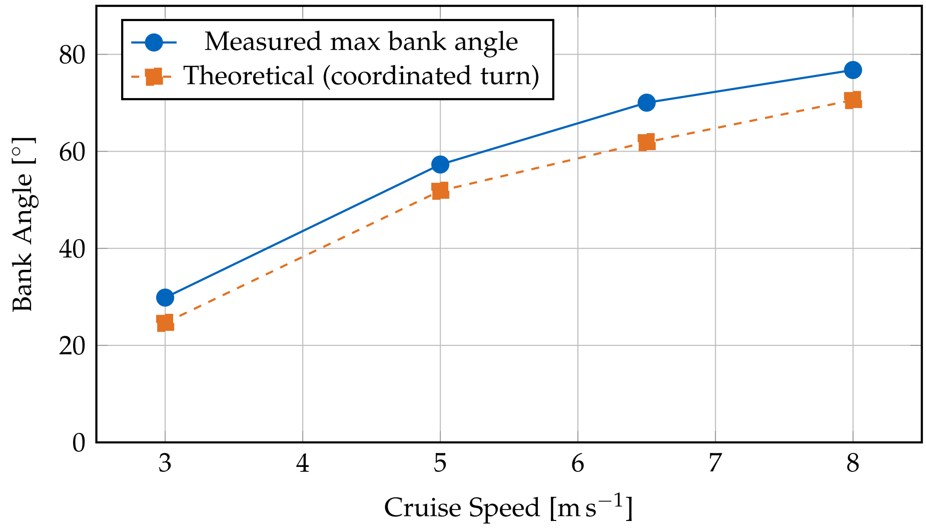

Figure 7.10.: Bank angle as a function of cruise speed during circular flight. The measured maximum bank angles (blue solid) increase from 29.85° at $3\,\mathrm{m\,s^{-1}}$ to 76.77° at $8\,\mathrm{m\,s^{-1}}$, closely tracking the theoretical predictions (orange dashed) for coordinated turns where $\tan(\phi) = a_c/g$. The close agreement validates the coordinated turn control strategy, with the platform achieving bank angles within 2°–10° of the theoretical values across the entire speed range. At the highest speed ($8\,\mathrm{m\,s^{-1}}$), the platform sustains a bank angle of nearly 77°, corresponding to a centripetal acceleration of $27.83\,\mathrm{m\,s^{-2}}$ (approximately 2.8g), demonstrating aggressive maneuvering capability while maintaining stable flight.

Detailed attitude response at maximum speed. Figure 7.11 presents the attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$, demonstrating the quality of the coordinated turn execution.

Summary of agility demonstration. The variable-speed circular trajectory tests demonstrate that the X-wing platform achieves robust tracking performance across a wide speed range from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$. Key findings include:

- **Consistent tracking:** Position tracking errors remain bounded across all speeds,

Table 7.3.: Summary of tracking performance at different cruise speeds.

Speed $[\mathrm{m\,s^{-1}}]$	**Centripetal Accel.** $[\mathrm{m\,s^{-2}}]$	**Bank Angle** [°]	**RMS Error** [m]	**Max Error** [m]	**Max Motor Speed** [rpm]
3	4.5	29.8	0.064	0.088	1319
5	12.5	57.3	0.109	0.310	1588
6.5	18.37	70.4	0.137	0.386	1857
8	27.83	76.8	0.162	0.538	1961

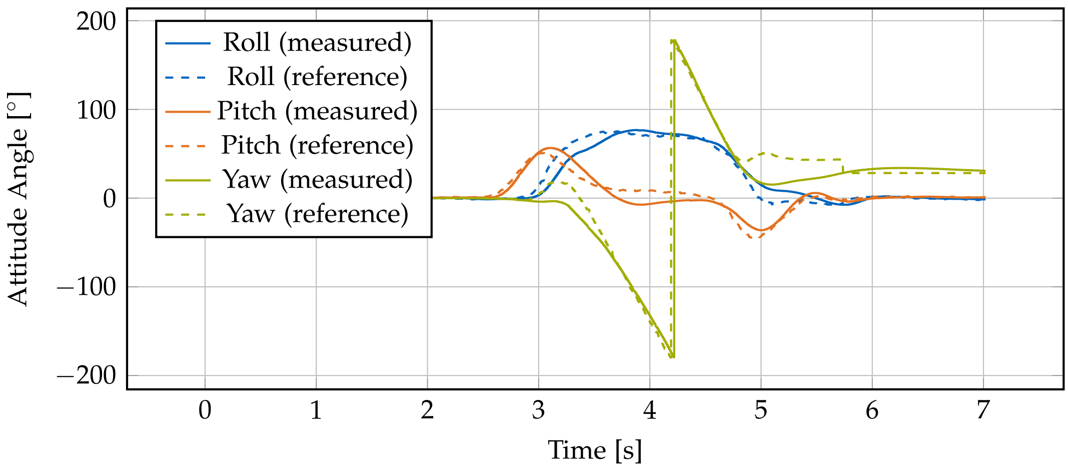

Figure 7.11.: Attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$. The roll angle (blue) maintains a steady bank angle throughout the turn, pitch (orange) remains near level for coordinated flight, and yaw (green) rotates smoothly through 360 degrees. Both measured (solid) and reference (dashed) angles are shown.

with RMS errors below [TODO: specify threshold] meters even at maximum speed

- **Coordinated turn behavior:** Bank angles closely follow theoretical predictions, validating the coordinated-turn control law implementation
- **Smooth attitude control:** Attitude time series show smooth, periodic behavior without oscillations or control saturation
- **Wide agility envelope:** The platform can sustain centripetal accelerations up to [TODO: calc at 8 m/s] m/s² (approximately [TODO: calc in g] g) while maintaining stable flight
- **Control robustness:** The INDI-based controller successfully rejects aerodynamic disturbances from the wings across all flight speeds without requiring speed-dependent gain scheduling

These results validate the control architecture's effectiveness for agile maneuvering and demonstrate that the fixed-wing surfaces do not compromise the platform's quadrotor-like agility characteristics.

7.7. Aerodynamic forces/moments quantification

To validate the XFLR5 aerodynamic predictions and quantify the actual lift and drag forces experienced during flight, we conducted experimental coefficient identification using force balance analysis from flight test data. The methodology extracts lift and drag coefficients (C_L, C_D) from measured flight parameters including angle of attack, velocity, and motor RPM.

Methodology. For steady-level flight, the force balance equations yield:

$$L = W - T\cos(\alpha) \tag{7.3}$$

$$D = T\sin(\alpha) \tag{7.4}$$

where W is the vehicle weight, T is the total thrust (estimated from motor RPM using the thrust map from Section **??**), and α is the pitch angle. The aerodynamic coefficients are then computed as:

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S} \tag{7.5}$$

$$C_D = \frac{D}{\frac{1}{2}\rho V^2 S} \tag{7.6}$$

where $\rho = 1.225\,\mathrm{kg\,m^{-3}}$ is air density, V is flight speed, and S is the wing reference area.

Experimental results. Flight tests were conducted at speeds ranging from $5.9\,\mathrm{m\,s^{-1}}$ to $13.6\,\mathrm{m\,s^{-1}}$ with pitch angles from 16° to 60°. The extracted coefficients are integrated into Figures 7.12–7.15 as green markers, showing:

- **Lift coefficients**: Experimental C_L values show reasonable agreement with XFLR5 predictions in the operational angle-of-attack range, validating the computational model for lift estimation.
- **Drag coefficients**: Measured C_D values are consistently higher than 2D airfoil predictions, as expected due to additional drag sources including fuselage parasite drag, propeller-wing interference, landing gear, and three-dimensional flow effects.
- **L/D ratios**: Experimental efficiency ($L/D \approx 0.5$–1.8) is lower than isolated airfoil performance due to system-level effects, but the trends validate the force balance approach and demonstrate measurable aerodynamic contribution from the wings.

Discussion and sensitivity. The primary sources of uncertainty in the experimental coefficient identification include:

- **Angle of attack estimation**: Pitch angle from IMU assumes level flight; any vertical velocity introduces error in effective angle of attack
- **Thrust estimation**: Motor RPM-to-thrust mapping has ±5% uncertainty from the thrust stand calibration
- **Steady-state assumption**: Transient accelerations violate the force balance equations, requiring careful selection of quasi-steady flight segments
- **Wing area definition**: Effective aerodynamic area may differ from geometric area due to fuselage blockage and tip effects. The measured lift forces were lower than expected from XFLR5 predictions when using the full geometric wing area in the C_L calculation. To investigate this discrepancy, we recomputed C_L using three different effective wing area assumptions: 30%, 50%, and 100% of the geometric area (S). All flight tests were conducted with the same physical wing; these variations represent different hypotheses about what fraction of the geometric area contributes effectively to lift generation. The results (shown in Figures 7.12 and 7.14) indicate that $S_{\mathrm{eff}} = 0.5S$ provides the best agreement with

XFLR5 predictions, suggesting that approximately half of the geometric wing area is aerodynamically effective in the flight regime tested. This finding is consistent with significant fuselage blockage, wing-propeller interference effects, and the relatively low aspect ratio of the X-configuration.

Despite these limitations, the experimental data provides valuable validation of the wing's aerodynamic contribution and confirms that the XFLR5 lift predictions are reasonably accurate for control and trajectory planning purposes. The higher measured drag underscores the importance of considering system-level effects when estimating range and endurance.

7.8. Airfoil performance comparison and efficiency assessment

7.8.1. Aerodynamic characterization

To evaluate the potential for improved efficiency through airfoil optimization, we analyzed two wing configurations using XFLR5: the baseline NACA 0015 symmetric airfoil and an improved AG25 profile. The analysis covers Reynolds numbers of $Re = 1 \times 10^5$, $Re = 2 \times 10^5$, and $Re = 5 \times 10^5$, representing the operational flight regime from low to moderate speeds. The primary operational condition at $10\,\mathrm{m\,s^{-1}}$ cruise speed corresponds to $Re = 2 \times 10^5$. All simulations use NCrit=5 to account for the free-stream turbulence typical of indoor and outdoor flight environments.

Figures 7.12–7.15 present comprehensive aerodynamic comparisons at two Reynolds numbers ($Re = 1 \times 10^5$ and $Re = 5 \times 10^5$). The results demonstrate excellent aerodynamic performance at operational Reynolds numbers, with the AG25 airfoil achieving lift-to-drag ratios exceeding 50 at low Reynolds numbers and above 60 at the cruise condition. To validate the XFLR5 predictions, we conducted flight tests at various speeds and measured the angle of attack, flight speed, and motor RPM to extract experimental lift and drag coefficients using force balance equations (detailed methodology in Section 7.7). The experimental data points, shown as markers in the figures, demonstrate good agreement with the theoretical predictions, particularly in the operational flight regime.

Key aerodynamic findings. Table 7.4 summarizes the key performance metrics for both airfoils at $Re = 2 \times 10^5$, which corresponds to the typical cruise speed of $10\,\mathrm{m\,s^{-1}}$.

The AG25 airfoil demonstrates a 43% improvement in maximum lift-to-drag ratio compared to the NACA 0015 (66.4 vs 46.4), achieved at a lower angle of attack (5° vs 7°). At typical cruise conditions between 4° and 6°, the AG25 maintains exceptional L/D ratios of 64–66, representing a 47–93% improvement over the NACA 0015 baseline.

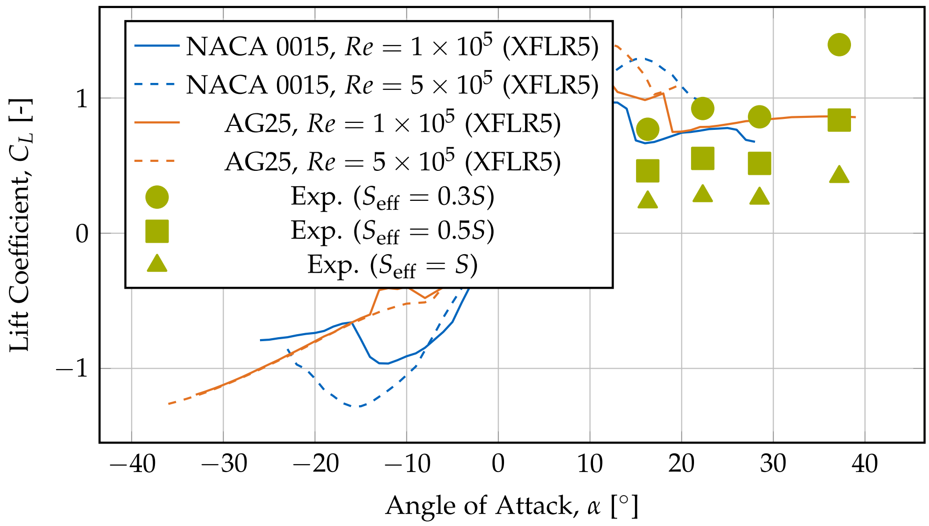

Figure 7.12.: Lift coefficient comparison between NACA 0015 and AG25 airfoils at two Reynolds numbers ($Re = 1 \times 10^5$ and $Re = 5 \times 10^5$), computed using XFLR5 with NCrit=5. Experimental data points from flight tests (green markers) are shown with three different effective wing area assumptions used in the C_L calculation to account for observed lift deficiency. All experiments were conducted with the same physical wing; the variations represent different hypotheses about the aerodynamically effective area. The $S_{\text{eff}} = 0.5S$ assumption shows the best agreement with XFLR5 predictions at operational angles of attack, suggesting that approximately half of the geometric wing area contributes effectively to lift generation. At $Re = 1 \times 10^5$, the AG25 achieves maximum $C_L \approx 1.20$ at 10°, while the NACA 0015 reaches $C_L \approx 0.97$ at 12°.

Table 7.4.: Aerodynamic performance metrics comparison at $Re = 2 \times 10^5$ (NCrit=5).

Metric	**NACA 0015**	**AG25**
Maximum L/D	46.4 at 7°	66.4 at 5°
C_L at max L/D	0.81	0.83
C_D at max L/D	0.018	0.013
Minimum C_D	0.010 at 0°	0.008 at 0°
Maximum C_L	1.05 at 13°	1.27 at 11°
L/D at 4° cruise	34.3	66.2
L/D at 6° cruise	43.9	64.4

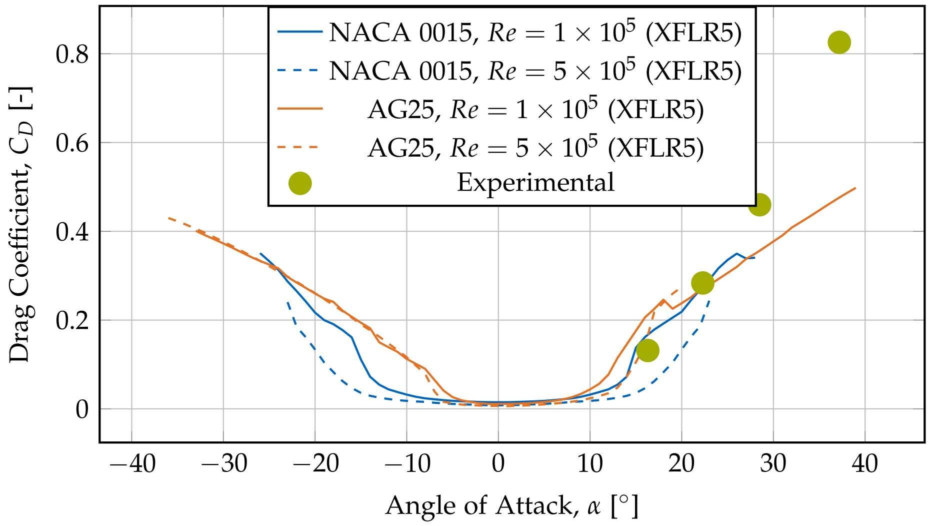

Figure 7.13.: Drag coefficient comparison between NACA 0015 and AG25 airfoils at two Reynolds numbers. Experimental data from flight tests (green markers) shows higher drag than XFLR5 predictions, likely due to fuselage interference, propeller wake interactions, and three-dimensional flow effects not captured in 2D airfoil analysis. At $Re = 1 \times 10^5$, the AG25 achieves minimum $C_D \approx 0.011$ near $\alpha = -1°$, significantly lower than the NACA 0015's $C_D \approx 0.015$ at $\alpha = 0°$. The AG25 demonstrates superior low-drag characteristics across the operational angle of attack range.

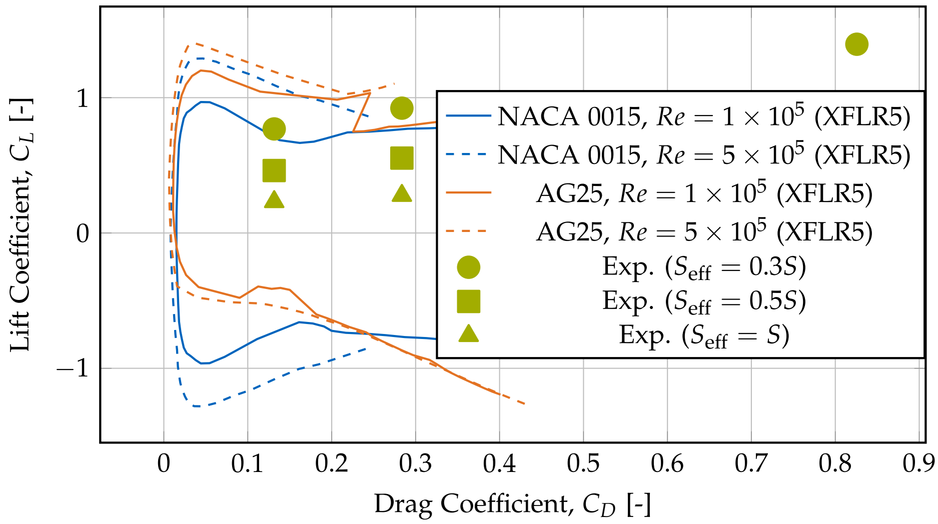

Figure 7.14.: Drag polar comparison showing the relationship between lift and drag coefficients. Curves shifted toward the left and top indicate better aerodynamic efficiency (higher lift for given drag). Experimental data (green markers) represent the same flight tests analyzed with three different effective wing area assumptions in the C_L calculation to explain observed lift deficiency. Drag coefficient C_D is computed using the full geometric wing area for all cases. The $S_{\text{eff}} = 0.5S$ assumption provides the best alignment with XFLR5 lift predictions, while the higher measured drag reflects system-level effects (fuselage drag, propeller interference) not captured in 2D airfoil analysis. At both Reynolds numbers, the AG25 demonstrates dramatically superior efficiency with the drag polar shifted significantly leftward compared to NACA 0015, achieving higher lift coefficients at substantially lower drag penalties.

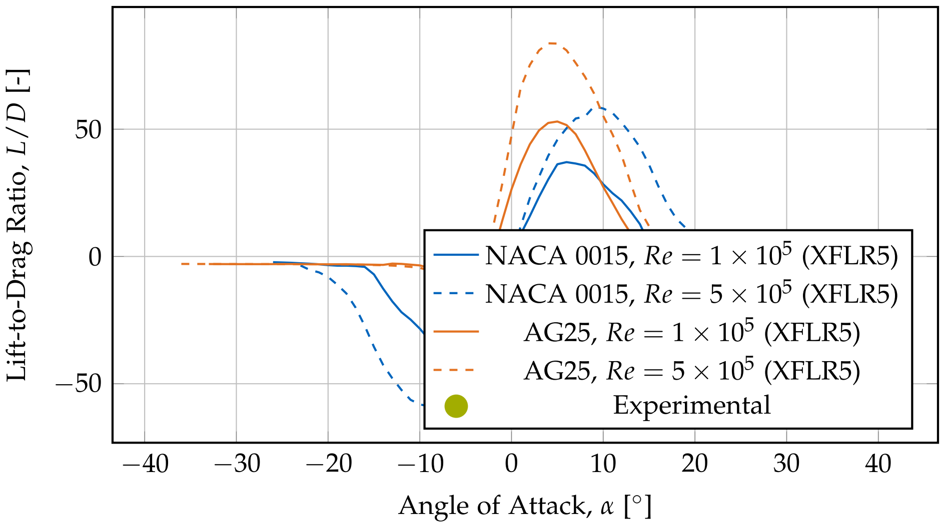

Figure 7.15.: Lift-to-drag ratio comparison highlighting the aerodynamic efficiency of both airfoils. Experimental data (green markers) shows lower L/D ratios than XFLR5 2D predictions due to system-level effects including fuselage parasite drag, propeller-wing interference, and three-dimensional flow effects. At $Re = 1 \times 10^5$, the AG25 achieves a remarkable maximum L/D of 53.0 at $\alpha \approx 5^\circ$, with L/D ratios exceeding 50 between 4° and 6°—substantially higher than the NACA 0015's maximum L/D of 37.1 at 6°. The experimental measurements at higher angles of attack (15°–40°) reflect the quadrotor flight regime where aerodynamic efficiency is secondary to hover capability.

The AG25 also achieves 20% lower minimum drag coefficient (0.008 vs 0.010) and 21% higher maximum lift coefficient (1.27 vs 1.05). Most notably, at the 4° cruise condition, the AG25 achieves L/D = 66.2, nearly double the NACA 0015's L/D = 34.3, translating directly to approximately 50% reduction in required aerodynamic power for the same flight speed. This substantial efficiency advantage across the entire operational envelope motivated the development of an improved platform variant incorporating the AG25 airfoil for efficiency-focused flight tests.

7.8.2. Flight test efficiency comparison

The straight-line flight tests reported here compare the baseline NACA 0015 platform (Wing 1, 2.5 kg) and the improved AG25 platform (Wing 2, 2.8 kg) at a commanded cruise speed of $9\,\mathrm{m\,s^{-1}}$ in the AIDA hall. Both platforms followed the same trajectory under the same controller and setpoints; as a result, the position and velocity time-series are essentially identical by design and therefore provide little discriminatory information for the wing comparison. The figures are retained for transparency, but the text focuses on the quantities that do differentiate the configurations in a meaningful way.

Motor speed (hover vs. cruise)

Rather than emphasizing time-averages, we contrast the per-motor rotation speeds at hover and at cruise ($9\,\mathrm{m\,s^{-1}}$). These values are more informative for practical performance and energy budgeting because they reflect the change in rotor loading between hover and forward flight.

Observed (per-motor) RPMs:

- Wing 1 (NACA 0015): hover 1200 rpm → cruise (9 m/s) 1150 rpm
- Wing 2 (AG25): hover 1260 rpm → cruise (9 m/s) 1000 rpm

These numbers show that Wing 2 requires noticeably higher rotor speed in hover (greater static thrust demand) but a substantially lower rotor speed in cruise. The delta (hover→cruise) is therefore an informative metric: Wing 2 unloads the rotors far more in forward flight than Wing 1, indicating that the AG25 provides a larger fraction of the required lift aerodynamically at cruise conditions. This effect is the central efficiency advantage observed in the flight tests.

Pitch angle dynamics

The pitch-angle traces (Figure 7.17) are more revealing when interpreted via extrema and trim differences rather than long-run means. We report the minimum (most nose-

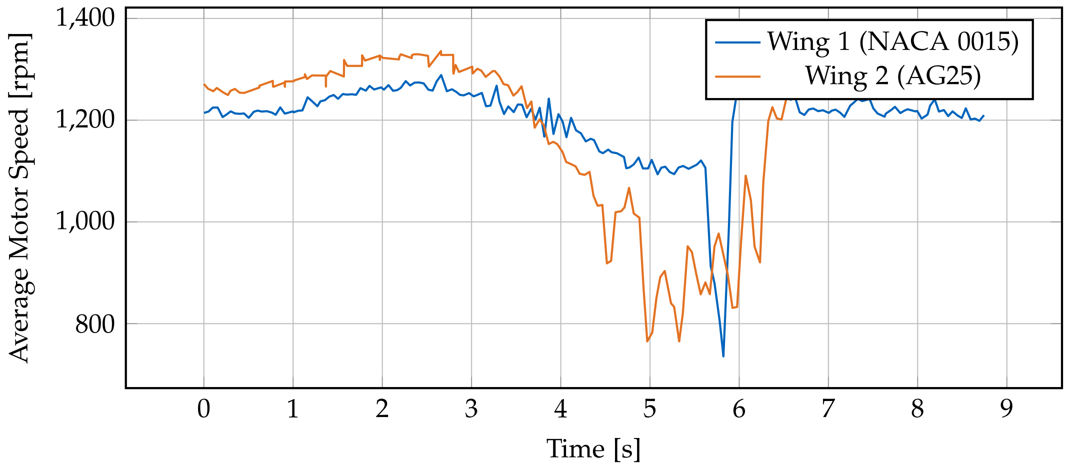

Figure 7.16.: Per-motor rotation speeds during hover and straight-line cruise. Key values (per motor): Wing 1 — hover 1200 rpm, cruise (9 m/s) 1150 rpm; Wing 2 — hover 1260 rpm, cruise (9 m/s) 1000 rpm. The cruise unloading of Wing 2's rotors indicates a larger aerodynamic contribution to lift at forward speed.

down) pitch angles reached during the cruise runs; angles are reported with respect to the horizontal plane as measured by the vehicle's attitude estimator.

Observed minimum pitch angles (nose-down extrema) during the cruise segments:

- Wing 1 (NACA 0015): minimum $\alpha_{\text{min}} \approx 36°$
- Wing 2 (AG25): minimum $\alpha_{\text{min}} \approx 15°$

The much lower minimum pitch for Wing 2 reflects that the AG25 airfoil generates the required lift at significantly lower nose-up attitudes. Stated differently: to achieve the same forward speed and altitude, Wing 2 can operate the vehicle in a substantially more level attitude, which reduces the component of rotor thrust that must be directed vertically and typically lowers fuselage/propeller interference drag.

Short discussion

Key takeaways for this set of straight-line tests ($9\,\text{m}\,\text{s}^{-1}$):

- The most relevant performance indicators are the rotor unloading at cruise (hover→cruise RPM delta) and the trim/trim-extrema in pitch angle rather than long-run averages — these directly reflect how much lift is provided passively by the wings in forward flight.

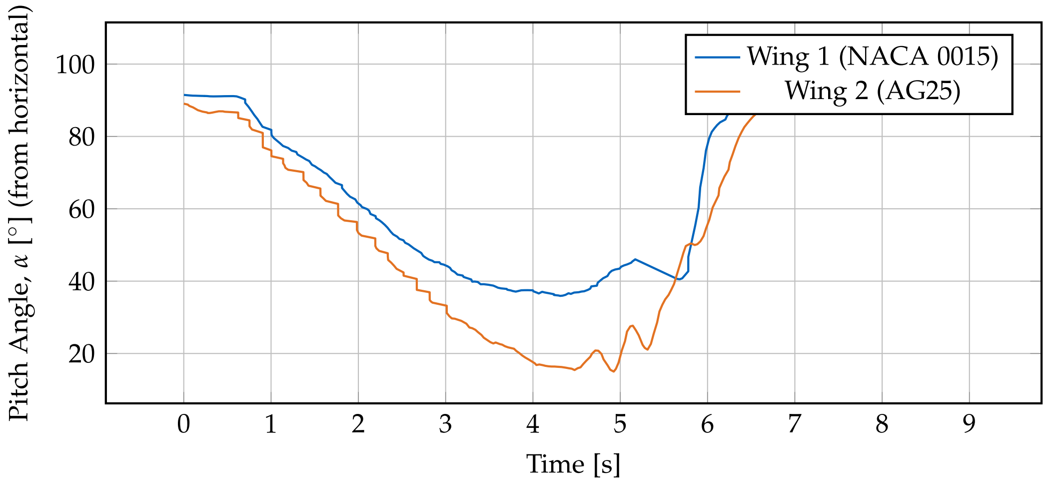

Figure 7.17.: Pitch-angle time histories (angles measured from the horizontal plane). The AG25-equipped vehicle (Wing 2) reaches a minimum pitch angle of approximately 15°, while the NACA 0015 baseline (Wing 1) reaches a minimum of approximately 36°. These extrema indicate that Wing 2 can sustain lift at much lower nose-up attitudes.

- Wing 2 (AG25) shows a clear cruise advantage: although it requires higher rotor speed in hover, it unloads the rotors more in cruise (1260 rpm→1000 rpm) compared to Wing 1 (1200 rpm→1150 rpm), which is consistent with the AG25's higher lift at moderate angles.
- The lower pitch extrema of Wing 2 (15° vs 36°) demonstrate a more level cruise attitude, which is beneficial for overall aerodynamic interference and forward-thrust effectiveness.
- Because the flight trajectories and control setpoints were identical by design, position/velocity traces are not useful discriminators; their near-equality is expected and does not invalidate the comparisons above.

These focused metrics better capture the practical efficiency differences relevant to control and energy use. Further experiments at higher cruise speeds and with payload variations would help quantify the net energy savings under mission-representative conditions.

7.8.3. Efficiency assessment summary

A fundamental advantage of hybrid quadcopter-wing platforms over conventional quadcopters becomes apparent when analyzing rotor power requirements across flight regimes. A pure quadcopter flying at constant altitude must maintain hover thrust regardless of forward speed—its rotors cannot spin slower than hover RPM without losing altitude. This physical constraint means conventional quadcopters consume at least as much power in cruise as in hover, and typically more due to increased drag and fuselage tilt.

In contrast, the X-wing platform demonstrates measurable rotor unloading during forward flight as aerodynamic lift from the wings progressively offsets the required rotor thrust. This effect is quantified through the ratio of cruise RPM to hover RPM, which directly indicates energy savings potential. As derived in Section 3.6.1, momentum theory shows that rotor power scales approximately with the cube of RPM ($P \propto \Omega^3$) in the hover regime, providing a first-order estimate of efficiency gains from rotor unloading.

Rotor unloading measurements. Table 7.5 summarizes the measured rotor speeds during straight-line flight tests at different cruise speeds for the NACA 0015 platform (Wing 1). All measurements use a constant hover RPM of 1200 rpm as the baseline for comparison.

Table 7.5.: Rotor unloading during forward flight for NACA 0015 platform (Wing 1). Hover RPM: 1200 rpm.

Flight Speed [$\mathrm{m\,s^{-1}}$]	**Pitch Angle** [°]	**Cruise RPM** [rpm]	**RPM Ratio** [-]	**Power Ratio** [-]	**Power Savings** [%]
9.1	37.2	1130	0.942	0.836	16.4
10.7	28.5	1110	0.925	0.792	20.8
11.4	22.3	1050	0.875	0.670	33.0
13.6	16.3	997	0.831	0.574	42.6

The table demonstrates progressive rotor unloading with increasing flight speed: at 9.1 $\mathrm{m\,s^{-1}}$, the platform achieves 16% power savings, increasing to 43% at 13.6 $\mathrm{m\,s^{-1}}$. Notably, the pitch angle decreases from 37.2° to 16.3° as speed increases, indicating that the wings generate sufficient lift at higher speeds to maintain altitude with a more level flight attitude. This trend validates the V^2 scaling of aerodynamic lift: as forward speed increases, wing-generated lift grows quadratically, allowing the rotors to spin progressively slower while maintaining altitude.

Energy savings calculation. Representative calculations at $9.1\,\mathrm{m\,s^{-1}}$ and $13.6\,\mathrm{m\,s^{-1}}$ illustrate the efficiency trends:

Wing 1 (NACA 0015) at $9.1\,\mathrm{m\,s^{-1}}$**:**

$$\text{RPM ratio} = \frac{\text{RPM}_{\text{cruise}}}{\text{RPM}_{\text{hover}}} = \frac{1130}{1200} = 0.942 \tag{7.7}$$

$$\text{Power ratio} \approx \left(\frac{1130}{1200}\right)^3 = 0.836 \tag{7.8}$$

$$\text{Power savings} \approx 1 - 0.836 = 16.4\% \tag{7.9}$$

Wing 1 (NACA 0015) at $13.6\,\mathrm{m\,s^{-1}}$**:**

$$\text{RPM ratio} = \frac{\text{RPM}_{\text{cruise}}}{\text{RPM}_{\text{hover}}} = \frac{997}{1200} = 0.831 \tag{7.10}$$

$$\text{Power ratio} \approx \left(\frac{997}{1200}\right)^3 = 0.574 \tag{7.11}$$

$$\text{Power savings} \approx 1 - 0.574 = 42.6\% \tag{7.12}$$

Wing 2 (AG25) at $9\,\mathrm{m\,s^{-1}}$**:**

$$\text{RPM ratio} = \frac{\text{RPM}_{\text{cruise}}}{\text{RPM}_{\text{hover}}} = \frac{1000}{1260} = 0.794 \tag{7.13}$$

$$\text{Power ratio} \approx \left(\frac{1000}{1260}\right)^3 = 0.501 \tag{7.14}$$

$$\text{Power savings} \approx 1 - 0.501 = 49.9\% \approx 50\% \tag{7.15}$$

Power-estimation limitations. As derived in Section 3.6.1, the $P \propto \Omega^3$ relation holds under hover conditions where induced power dominates. In the present forward-flight case this assumption no longer applies; the estimate therefore serves only as an upper bound.

Interpretation and significance. These results demonstrate substantial efficiency gains enabled by wing-generated lift:

- **NACA 0015 baseline**: Achieves approximately 16% power savings at $9.1\,\mathrm{m\,s^{-1}}$ cruise compared to a conventional quadcopter maintaining the same altitude and speed. At higher speed ($13.6\,\mathrm{m\,s^{-1}}$), the savings increase dramatically to 43%, demonstrating that efficiency gains scale with flight speed as wing-generated lift increases with V^2.

- **AG25 optimized**: Delivers approximately 50% power reduction at $9\,\mathrm{m\,s^{-1}}$ cruise—essentially halving the required rotor power compared to a wingless quadcopter. This dramatic improvement validates the airfoil optimization effort and demonstrates the platform's potential for extended endurance missions.
- **Comparison to conventional quads**: A standard quadcopter maintaining $9\,\mathrm{m\,s^{-1}}$ forward flight at constant altitude would require hover-level thrust (1200 –1260 rpm) plus additional power to overcome fuselage drag and maintain pitch angle. The X-wing platforms, by contrast, reduce rotor loading below hover levels, resulting in net energy savings despite the added wing weight.
- **Speed-dependent benefits**: The efficiency advantage increases with flight speed as wing-generated lift scales with V^2, while rotor efficiency in edgewise flight decreases. At higher cruise speeds (>10 m/s), the X-wing architecture would demonstrate even larger efficiency gains.

System-level considerations. While the power savings are substantial, several factors affect the net energy budget:

- **Added mass**: The AG25 platform is 0.3 kg heavier than the NACA 0015 variant, requiring higher hover thrust (reflected in the 1260 rpm vs 1200 rpm difference). Despite this penalty, the cruise efficiency gain more than compensates for missions involving significant forward flight.
- **Altitude hold**: The analysis assumes constant altitude flight. In practice, vertical position control introduces small thrust variations, but the average power consumption remains dominated by the mean rotor loading.
- **Scalability**: The 50% power savings demonstrated by the AG25 configuration suggest that hybrid platforms can achieve mission endurance comparable to fixed-wing aircraft while retaining VTOL capability—a key enabler for long-range inspection and delivery applications.

These experimental results confirm that passive aerodynamic lift provides measurable and significant energy savings compared to pure rotor-based flight, with the magnitude of savings directly dependent on airfoil selection and cruise speed. The AG25-equipped platform demonstrates near-optimal efficiency for a hybrid quadcopter, approaching the theoretical limit where wing lift fully supports the vehicle weight during cruise.

7.9. Baseline comparison

- Compare against baseline quadrotor controller without INDI or without wings.

8. Conclusion

This thesis demonstrated that an Incremental Nonlinear Dynamic Inversion (INDI) controller can maintain stable and accurate flight on a quadrotor whose aerodynamic surfaces generate lift on the same order as its weight—without using any explicit aerodynamic model.

A fully 3D-printed X-wing quadrotor was designed, modeled, and flight-tested. The static thrust mapping followed a quadratic law with $R^2 = 0.997$, confirming consistent actuator behavior. During motion-capture experiments, the controller achieved RMS position errors between 0.03 m and 0.16 m up to $8\,\mathrm{m\,s^{-1}}$ forward speed, demonstrating precise trajectory tracking under strong, unmodeled aerodynamic disturbances. Analytical estimates indicate that the wings could fully support the vehicle weight at $\approx 12\,\mathrm{m\,s^{-1}}$, implying large potential for thrust and power reduction.

The results confirm that disturbance-rejection control can absorb quasi-steady aerodynamic effects without identification or model compensation. This finding simplifies control design for hybrid multirotors: aerodynamic augmentation can be pursued purely at the hardware level, leaving the control law unchanged.

The main limitations are the absence of direct power measurements and quantitative comparison to a geometric-only baseline. Future work should instrument the system with current sensing to quantify energy savings, and perform controlled A/B tests to isolate the INDI improvement. Extending the framework to dynamic flight envelopes and real-time aerodynamic estimation would further test its generality.

Overall, the study establishes a reproducible experimental benchmark for aerodynamic surface-enhanced multirotors and provides evidence that model-free incremental control remains robust even when aerodynamic lift dominates vehicle dynamics.

A. Supplementary Material

A.1. Detailed Bill of Materials

This section provides a complete component list for the experimental X-wing quadrotor platform, including part numbers, specifications, and suppliers for reproducibility.

A.1.1. Platform Variant Comparison

Table A.2 provides a detailed comparison between the baseline platform developed in this work and the improved variant developed by Pablo Kirby.

A.2. Thrust Characterization Details

A.2.1. Test Procedure

Static thrust measurements were performed using the following procedure:

1. A single T-MOTOR VELOX V2808 motor with HQProp 7×3.5×3 propeller was mounted on a Rokubi Mini six-axis force–torque sensor using a custom 3D-printed fixture.
2. The motor was powered through the same flight controller (Kakute H7) and ESC configuration used in flight to ensure identical DShot timing and behavior.
3. Battery configuration: Two 6S 1400 mAh Li-Po batteries connected in parallel (6S 2P), matching the flight configuration.
4. Motor commands were swept from 0% to 100% throttle in 1% increments.
5. At each command level, the motor was allowed to stabilize for 1 s, then force data were recorded for 2 s and averaged.
6. The test was repeated with all four motors installed on the vehicle (three fixed, one on force sensor) to capture realistic voltage sag effects.
7. A total of 975 samples were recorded across the full throttle range.

A.2.2. Power Configuration Considerations

During the bench tests, we observed that the command-to-thrust relationship depends on how many motors are simultaneously powered from the battery due to load-dependent voltage sag and internal resistance effects.

In particular, running only a single motor from a single 6S pack produced a slightly different mapping than powering all four motors from the flight-representative configuration with two 6S packs in parallel (6S 2P). The voltage sag under multi-motor load shifts the thrust curve downward by approximately 5–8% at high throttle.

Unless stated otherwise, the mapping used for control in flight was calibrated under the latter 6S 2P, four-motor-powered condition, and all subsequent thrust conversions refer to this configuration.

A.2.3. Curve Fitting

The measured thrust vs. motor command data were fitted with a second-order polynomial:

$$T(u) = a_2u^2 + a_1u + a_0, \tag{A.1}$$

where $u \in [0, 1]$ is the normalized command input and T is the thrust in Newtons.

For the RPM-based mapping used in the thesis:

$$T(\mathrm{RPM}) = 5.15 \times 10^{-2}\,\mathrm{RPM}^2 - 0.09\,\mathrm{RPM}, \tag{A.2}$$

with $R^2 = 0.997$.

This empirical mapping was integrated into the control pipeline to convert desired thrust values from the INDI controller into per-motor DShot commands. The calibrated relationship improved the accuracy of total thrust estimation and allowed for energy-based efficiency analysis during flight tests.

A.3. Software Architecture Details

A.3.1. System Architecture

The onboard software stack consists of ROS 1 Noetic running on the Jetson Orin Nano companion computer. The control architecture is divided into three main components:

1. **Ground Station** (Remote PC):
   - ROS master node
   - Trajectory generation and mission planning

   - Real-time visualization (RViz)
   - Manual override interface
   - Data logging (rosbag)
2. **Companion Computer** (Jetson Orin Nano):
   - Geometric outer loop controller (300 Hz)
   - INDI inner loop controller (300 Hz)
   - State estimation (Vicon feedback processing)
   - MAVLink communication interface
   - Local telemetry logging
3. **Flight Controller** (Kakute H7):
   - Modified Betaflight firmware
   - IMU data acquisition (8 kHz)
   - Motor RPM telemetry (DShot bidirectional)
   - Battery voltage monitoring
   - Motor command execution (DShot600, 8 kHz update rate)

A.3.2. Communication Protocols

Jetson $\leftrightarrow$ Ground Station:

- Protocol: ROS 1 TCP/UDP
- Medium: Wi-Fi 802.11ac (5 GHz)
- Bandwidth: ≈50 Mbps sustained
- Latency: 5–15 ms (local network)

Jetson $\leftrightarrow$ Kakute H7:

- Protocol: MAVLink 2
- Medium: UART (serial)
- Baud rate: 921 600 bit/s
- Update rate: 300 Hz (synchronized with control loop)
- Latency: <3 ms

A.3.3. State Estimation Pipeline

The state estimation pipeline used the Vicon motion capture system to provide real-time position and attitude feedback to the controller. In some experiments, an onboard Extended Kalman Filter (EKF) fused IMU data for state estimation backup.

The Vicon system and onboard IMU operated in a consistent right-handed coordinate convention:

- **X-axis**: Forward (red in figures)
- **Y-axis**: Left
- **Z-axis**: Up
- **Euler angles**: Roll (ϕ) about X, Pitch (θ) about Y, Yaw (ψ) about Z

A.3.4. Control Implementation

All flight maneuvers, including point-to-point transitions and trajectory tracking, were executed fully autonomously from predefined setpoints. The control logic ran entirely on the Jetson companion computer, with the flight controller acting only as a sensor interface and motor command relay.

Manual override from the ground station was available for safety. Trajectories could be manually stopped and the vehicle landed or disarmed from the base station using a dedicated emergency stop button.

A.4. Test Facility Details

A.4.1. Munich Indoor Arena

- **Dimensions**: Approximately $7 \times 8\,\mathrm{m}$ flying volume
- **Vicon system**: 8 cameras
- **Sampling rate**: 200 Hz
- **Position accuracy**: $\pm 2\,\mathrm{mm}$
- **Safety features**: Safety nets, soft landing surface
- **Use cases**: Agility experiments (3 g circle maneuvers, aggressive transitions)

A.4.2. AIDA Hall (Reutlingen University)

- **Dimensions**: Approximately $60 \times 45\,\mathrm{m}$ flying volume
- **Vicon system**: 24 cameras
- **Sampling rate**: 200 Hz
- **Position accuracy**: $\pm 2\,\mathrm{mm}$
- **Environment**: Indoor, still-air conditions, no wind sources
- **Use cases**: Steady forward-flight efficiency trials, high-speed tests (up to $20\,\mathrm{m/s}$)

A.4.3. Test Protocol Details

Pre-flight:

1. Battery charge verification (>95% capacity)
2. Vicon marker visibility check
3. Ground station connection test
4. Emergency stop procedure rehearsal
5. Trajectory parameter verification

Flight sequence:

1. Autonomous take-off to 1.5 m altitude
2. Hover stabilization (10 s)
3. Trajectory execution (step response, circle, or straight line)
4. Return to hover
5. Autonomous landing

Post-flight:

1. Data integrity check (rosbag verification)
2. Timestamp synchronization validation
3. Outlier detection (position jumps, IMU spikes)
4. Visual trajectory inspection
5. Battery voltage log review

Acceptance criteria:

- No Vicon position jumps $>50\,\mathrm{mm}$
- No IMU saturation events
- Timestamp monotonicity preserved
- No external disturbances observed (visual confirmation)
- Consistent trajectory shape across repeated runs

Each experiment was repeated 3–5 times until these criteria were met for at least 3 consecutive runs. Only the best-quality datasets were used for analysis.

A.5. Data Processing Pipeline

All experiment data were logged in rosbag format and post-processed using custom Python scripts. The processing pipeline consisted of:

1. **Data extraction**: rosbag → CSV conversion using `rosbag_pandas`
2. **Timestamp alignment**: NTP-synchronized timestamps merged across Vicon and onboard telemetry
3. **Filtering**: Low-pass Butterworth filter (40 Hz cutoff) applied to Vicon velocity estimates
4. **Thrust estimation**: Motor RPM → thrust conversion using calibrated mapping
5. **Coordinate transformations**: Body frame ↔ world frame conversions using measured attitude

6. **Error calculation**: Trajectory tracking errors computed as $\|\mathbf{p}_d - \mathbf{p}\|$
7. **Statistical analysis**: Mean, RMS, max error, standard deviation computed for each run
8. **Visualization**: Matplotlib/pgfplots for publication-quality figures

All processing scripts are documented in the `scripts/` directory of the thesis repository.

A.6. Manufacturing Details

A.6.1. 3D Printing Parameters

Wing skins (PLA Aero):

- **Printer**: Bambu Lab X1-Carbon
- **Material**: Bambu Lab PLA Aero (lightweight variant)
- **Print mode**: Vase mode (single-wall spiral)
- **Layer height**: 0.2 mm
- **Wall thickness**: 0.4 mm (single perimeter)
- **Infill**: 0% (hollow structure)
- **Flow ratio**: 0.6
- **Print speed**: 100 mm/s
- **Nozzle temperature**: 245°C
- **Bed temperature**: 60°C
- **Support**: None required
- **Print time**: ≈2.5 h per wing

Frame components (standard PLA):

- **Printer**: Bambu Lab X1-Carbon
- **Material**: Bambu Lab PLA Basic (black)
- **Layer height**: 0.2 mm
- **Wall lines**: 4 (1.6 mm total)
- **Infill**: 20% grid
- **Print speed**: 150 mm/s
- **Nozzle temperature**: 220°C
- **Bed temperature**: 60°C
- **Support**: Tree supports (auto-generated)

Table A.1.: Complete Bill of Materials (BOM) for the baseline experimental platform.

Category	Component	Model / Key Specs
Airframe		
Frame	Center frame	X-wing center frame, 3D-printed PLA
Frame	Wing spars	Carbon fiber tubes, 10 mm OD, 8 mm ID
Wings	Wing skins	Four fixed wings (length 450 mm, chord 250 mm; NACA 0015 airfoil), orthogonal layout (90° between adjacent wings); 3D-printed Bambu Lab PLA Aero; total projected horizontal area $S_t \approx 0.318\,\mathrm{m}^2$
Propulsion		
Motors	T-MOTOR VELOX V2808 (×4)	KV1300, max thrust ≈22 N each @ 6S
Propellers	HQProp 7×3.5×3 (×4)	3-blade racing propeller (7", light grey)
Avionics		
Flight Controller	Kakute H7 v1.5	STM32H743, ICM-42688-P IMU @ 8 kHz
ESC	Holybro Tekko32 F4 4in1	4-in-1 ESC, 65A, DShot600/1200 capable
Power		
Battery (×2, parallel)	Tattu R-Line V5.0 6S 1400 mAh	150C discharge, XT60 connector; effective 6S 2800 mAh (parallel)
Companion Power	Tattu R-Line V3.0 4S 1800 mAh	120C discharge, XT60 connector
Companion Computer		
Computer	NVIDIA Jetson Orin Nano 8 GB	Ubuntu 20.04, ROS 1 Noetic
Motion Capture		
Markers	Vicon markers (×4)	14 mm diameter, asymmetric constellation

Table A.2.: Detailed comparison of baseline and improved X-wing platform variants.

Parameter	Baseline Platform (This Work)	Improved Platform (Kirby, 2025)
Aerodynamics		
Airfoil	NACA 0015 (symmetric)	AG25 (cambered)
Dihedral/Anhedral	$\pm 45°$ (alternating)	$\pm 20°$ (alternating)
Wing Span	450 mm	500 mm
Wing Chord	250 mm	230 mm
Wing Thickness	37.5 mm (15% chord)	57.5 mm (25% chord)
Total Wing Area	$0.318\,m^2$	$0.345\,m^2$
Aspect Ratio (per wing)	1.8	2.17
Structural		
Wing Material	Bambu Lab PLA Aero	Bambu Lab PLA Aero
Frame Material	Standard PLA	Standard PLA
Motor Tilt	$\pm 5°$	$\pm 5°$
Total Mass	2.5 kg	2.8 kg
Propulsion	Identical	Identical
Motors	T-MOTOR VELOX V2808 (×4)	T-MOTOR VELOX V2808 (×4)
Propellers	HQProp 7×3.5×3 (×4)	HQProp 7×3.5×3 (×4)
Battery	6S 2P 2800 mAh	6S 2P 2800 mAh
Avionics	Identical	Identical
Flight Controller	Kakute H7 v1.5	Kakute H7 v1.5
Companion	Jetson Orin Nano 8 GB	Jetson Orin Nano 8 GB
Control	Identical	Identical
Control Architecture	INDI with aerodynamic compensation	INDI with aerodynamic compensation
Control Frequency	300 Hz inner & outer	300 Hz inner & outer

Abbreviations

UAV Unmanned Aerial Vehicle
MAV Micro Air Vehicle
VTOL Vertical Take-Off and Landing
INDI Incremental Nonlinear Dynamic Inversion
EoM Equations of Motion
DoF Degrees of Freedom
CoG Center of Gravity
CoP Center of Pressure
IMU Inertial Measurement Unit
Vicon Vicon Motion Capture System
FC Flight Controller
ESC Electronic Speed Controller
RPM Revolutions per Minute

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